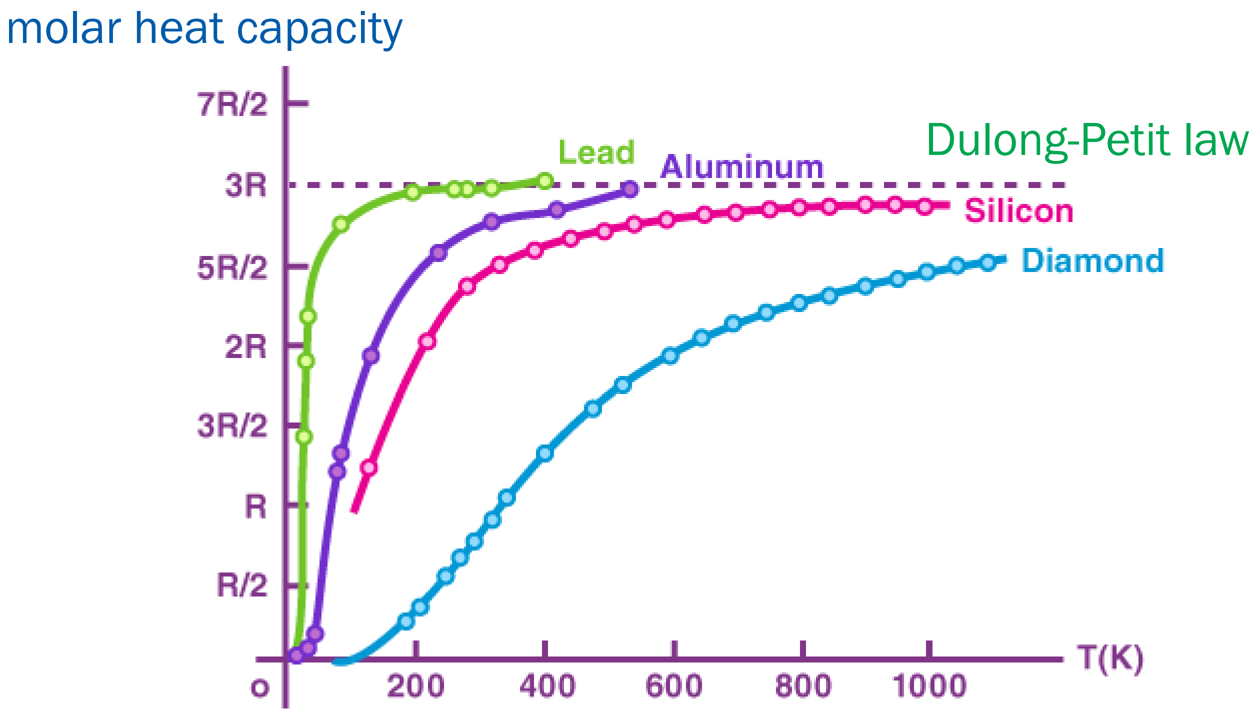
Physical Chemistry of Materials

# Lecture 13

## Electronic Structures of Solids

# Heat Capacity of Solid




(Image: https://blog.naver.com/jinest/70153707595)

Last Class

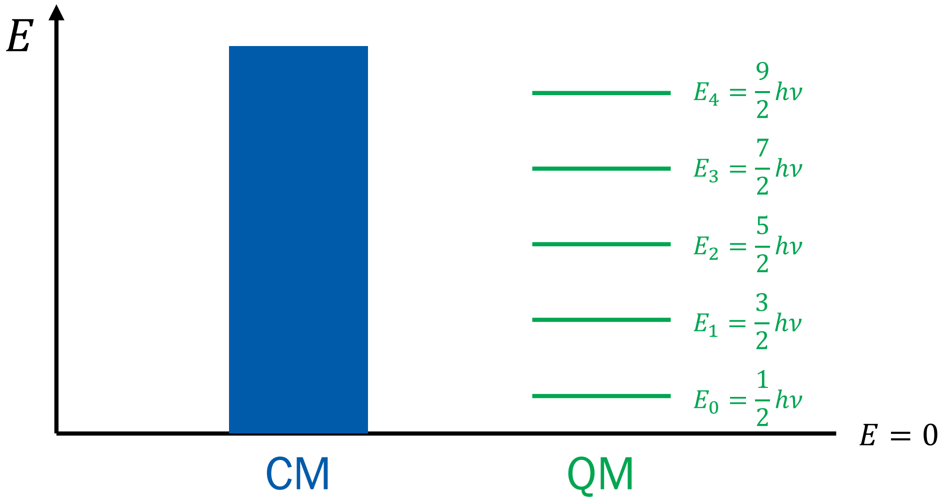
# Stat Mech of Crystal Vibration

| | Identical oscillators | Normal modes |
|---|---|---|
| Classical | Dulong-Petit | Dulong-Petit |
| Quantum | Einstein model | Debye model |

# "Phonon"

A phonon is a discrete quantum of vibration

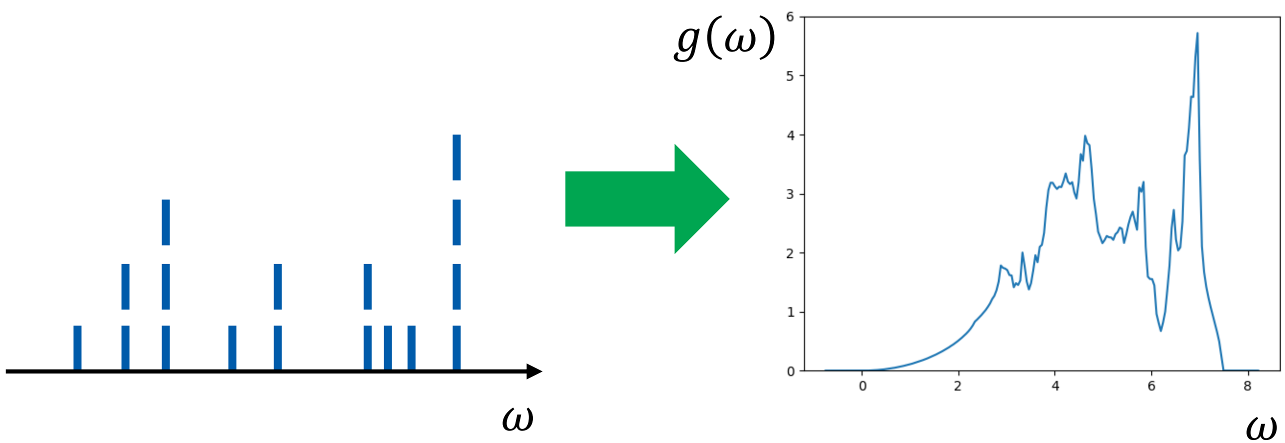
(bosonic)



Last Class

# Debye Model

- $3N$ independent (different) harmonic oscillators
- We have large $N$: many normal modes with different $\omega_i$
- Consider them as continuous!
- Density of states (DOS) $g(\omega)$



# DOS Calculation

- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$
- $h(k)\,\mathrm{d}k = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[k, k + \mathrm{d}k]$

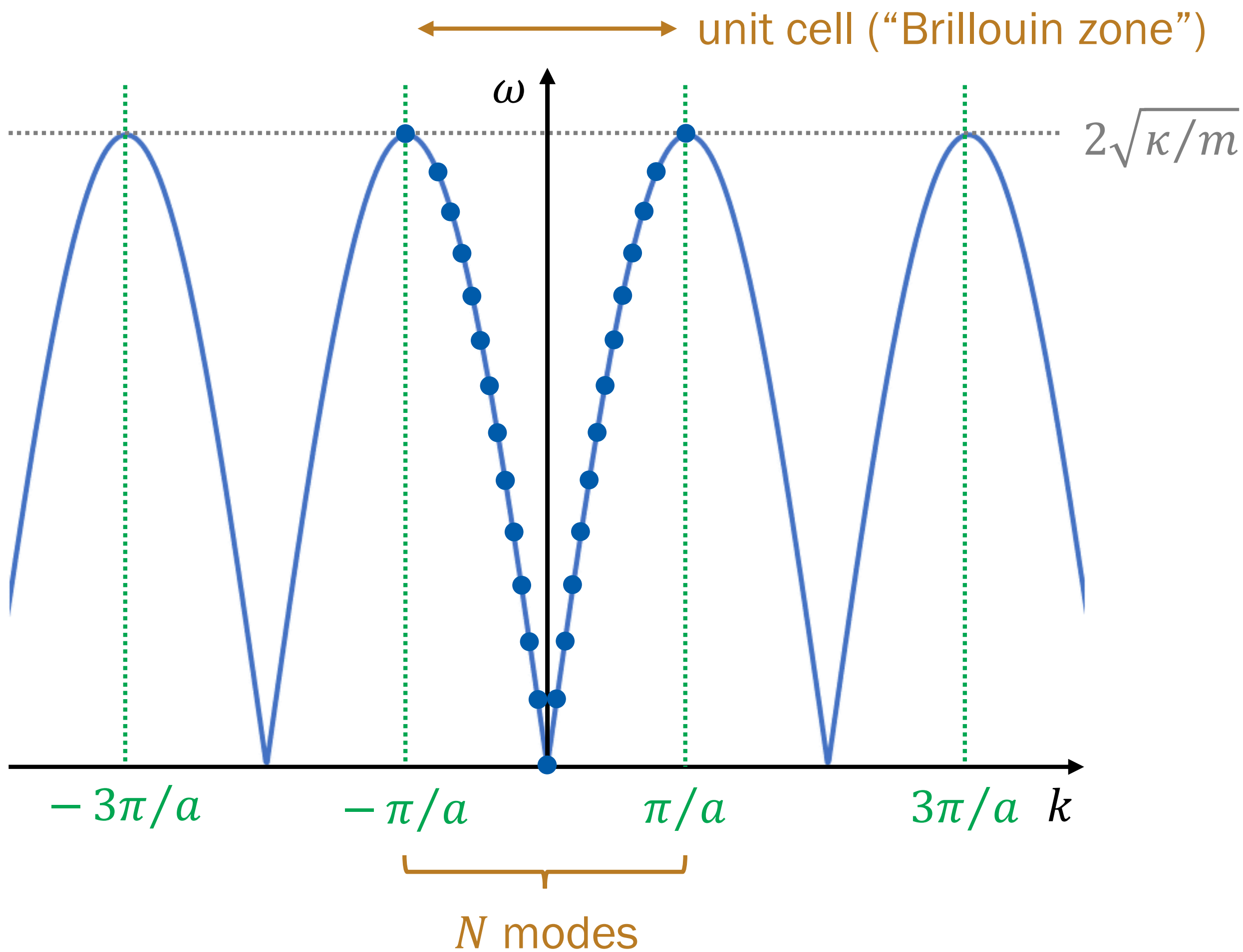
$$h(k) = \frac{\mathrm{d}N_{\mathrm{modes}}}{\mathrm{d}k}$$

How can we count the number of modes?

The dispersion relation is continuous only when $N \to \infty$; consider finite $N$ (= finite system size)!

Last Class

# Case of 1D Simple Model



Last Class

# DOS Calculation

- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$
- $h(k)\,\mathrm{d}k = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[k, k + \mathrm{d}k]$

$$h(k) = \frac{\mathrm{d}N_{\mathrm{modes}}}{\mathrm{d}k}$$

$$h_{1\mathrm{D}}(k) \sim \frac{N}{2\pi/a} = \frac{L}{2\pi}$$

$$h_{3\mathrm{D}}(\vec{k}) \sim \frac{N^3}{(2\pi/a)^3} = \frac{L^3}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

# DOS Calculation

- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$
- $h(k)\,\mathrm{d}k = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[k, k + \mathrm{d}k]$

$$h_{3\mathrm{D}}(\vec{k}) \sim \frac{N^3}{(2\pi/a)^3} = \frac{L^3}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

The number of modes in $[0, k]$ is then

$$N_{\mathrm{modes}}(k) = \int_{\mathrm{volume}} h_{3\mathrm{D}}(\vec{k})\,\mathrm{d}\vec{k} = \int_0^k h_{3\mathrm{D}}(\vec{k})\,4\pi k^2 \mathrm{d}k$$

$$N_{\mathrm{modes}}(k) = \frac{Vk^3}{6\pi^2}$$

# DOS Calculation

- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$

The number of modes in $[0, k]$ is

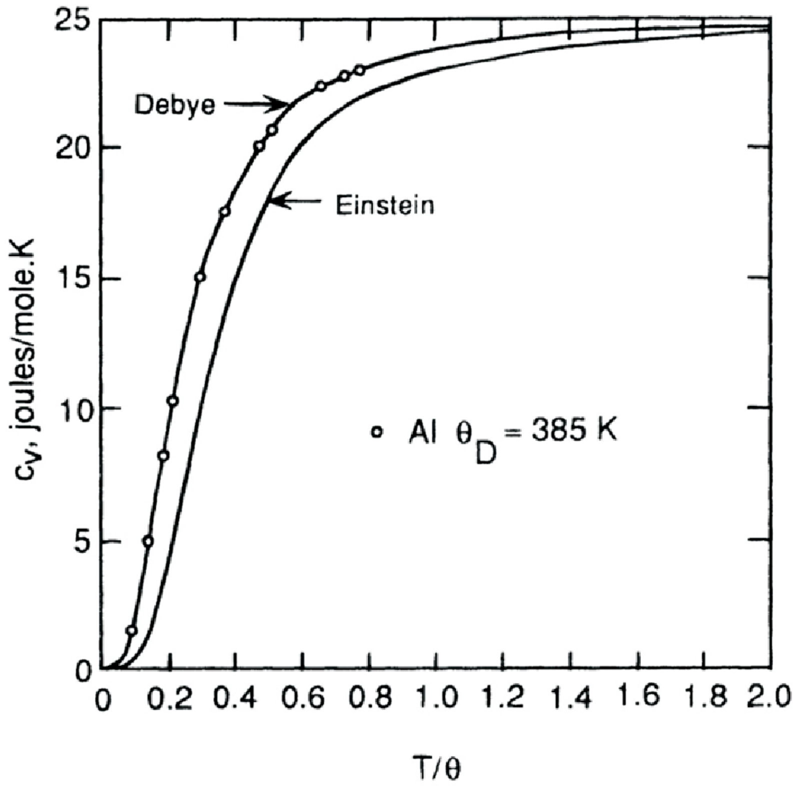
$$N_{\mathrm{modes}}(k) = \frac{Vk^3}{6\pi^2}$$

$$g(\omega) = \frac{\mathrm{d}N_{\mathrm{modes}}}{\mathrm{d}\omega} = \frac{Vk^2}{2\pi^2}\frac{1}{\mathrm{d}\omega/\mathrm{d}k}$$

For a simple dispersion relation, $\omega = vk$,

$$g(\omega) = \frac{V\omega^2}{2\pi^2 v^3}$$

# Einstein vs. Debye

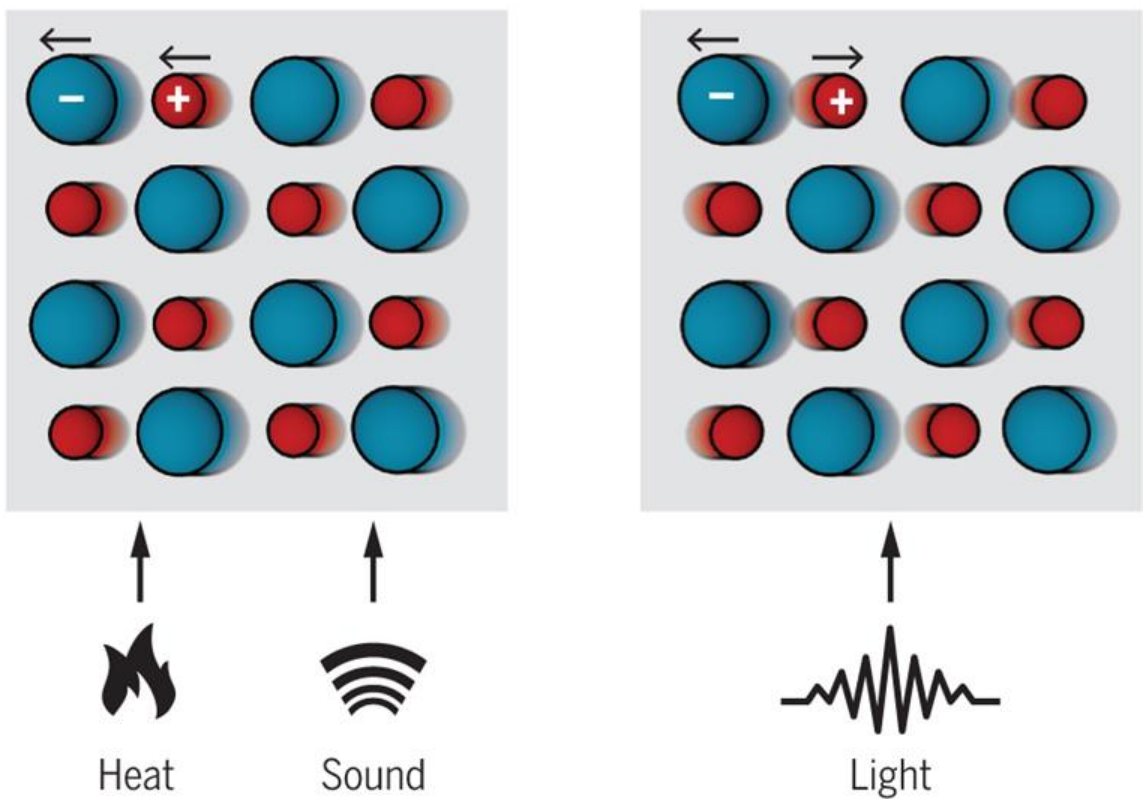



(Image: https://www.tf.uni-kiel.de/matwis/amat/td_kin_i/kap_1/backbone/r_su4.html)

# Two Types of Phonons

Acoustic

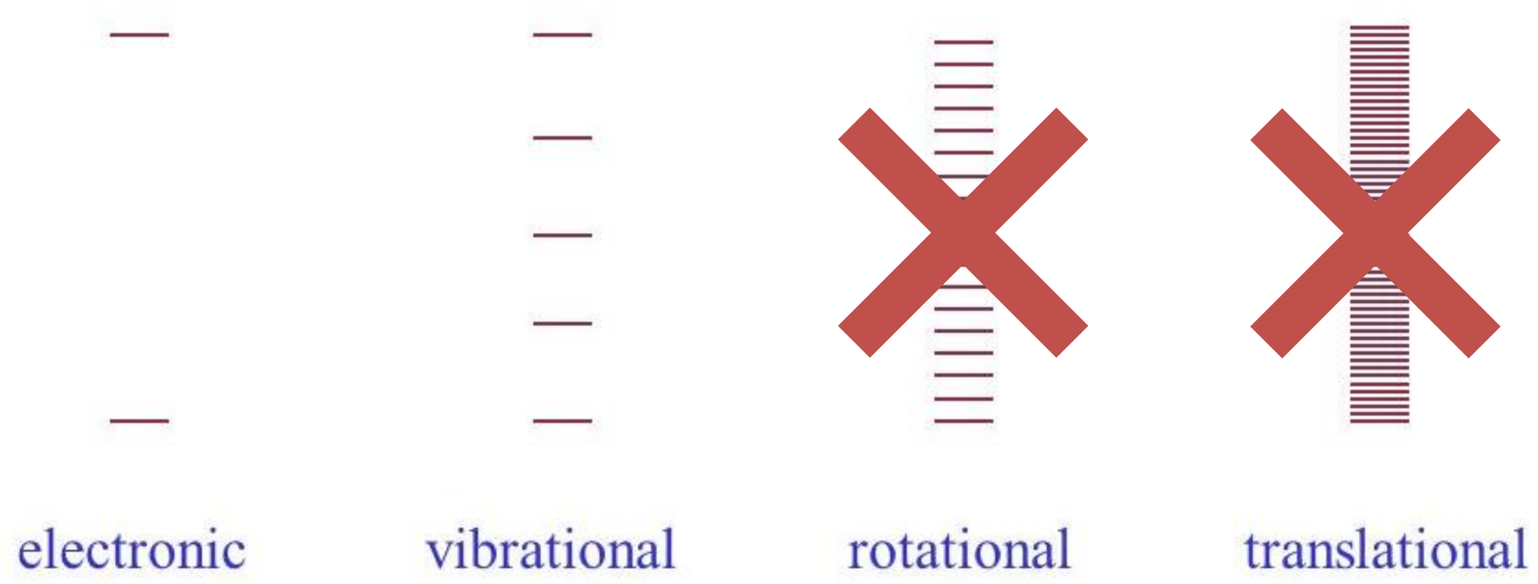
Optical




(Image: Juraschek and Spaldin, *Science* (2017))

# Crystal Energies

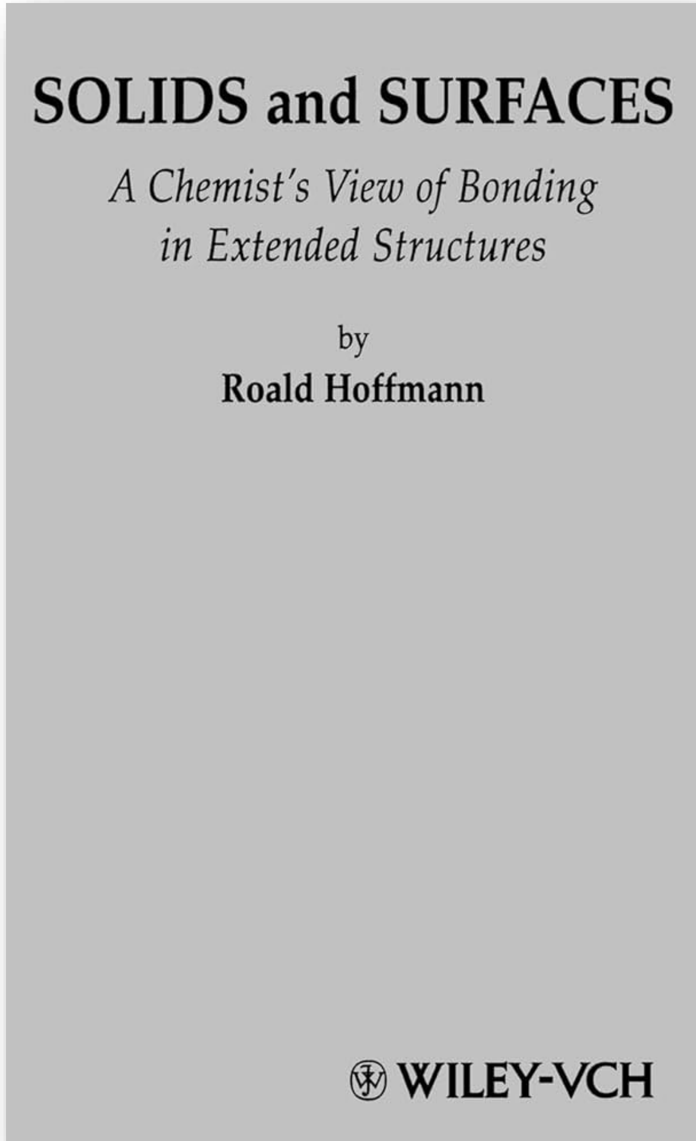
Molecular case:



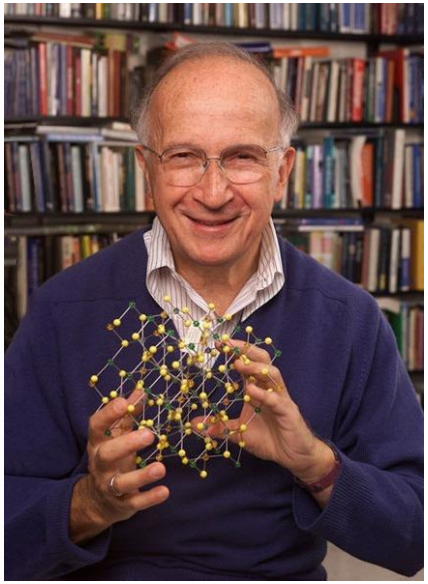

(Image: https://slideplayer.com/slide/6282680/)

내가 대학원생이었을 때(1970년대 중반) 실리콘 밸리에는 반도체 물리학자들이 결정성을 숭배하면서 평화롭게 살고 있다고 들었다. 결정성의 딸들인 원자가 띠(valence band)와 전도성 띠(conduction band)가 트랜지스터를 작동하고 번성하게 해준다는 것이다. 그러나 그들은 결정이 아니라 분자를 숭배하고, 그 자손인 LUMO와 HOMO가 트랜지스터를 움직이는 진짜 이유라고 믿고, 과거의 신을 믿는 사람들은 열등하고 순수하지 않다고 생각하는 화학자들에 의해서 침략을 당했다. 두 종족들은 엉터리 정보와 지저분한 음모와 상대방 종족이 숭배하는 신의 이름을 부르는 것마저 거절하면서 피나는 싸움을 벌였다. 서로 상대방의 연구비를 말려서 소멸시켜버리려고 했다. 전쟁은 교착 상태에 빠져버렸지만 그 흔적은 지금까지도 남아 있다.

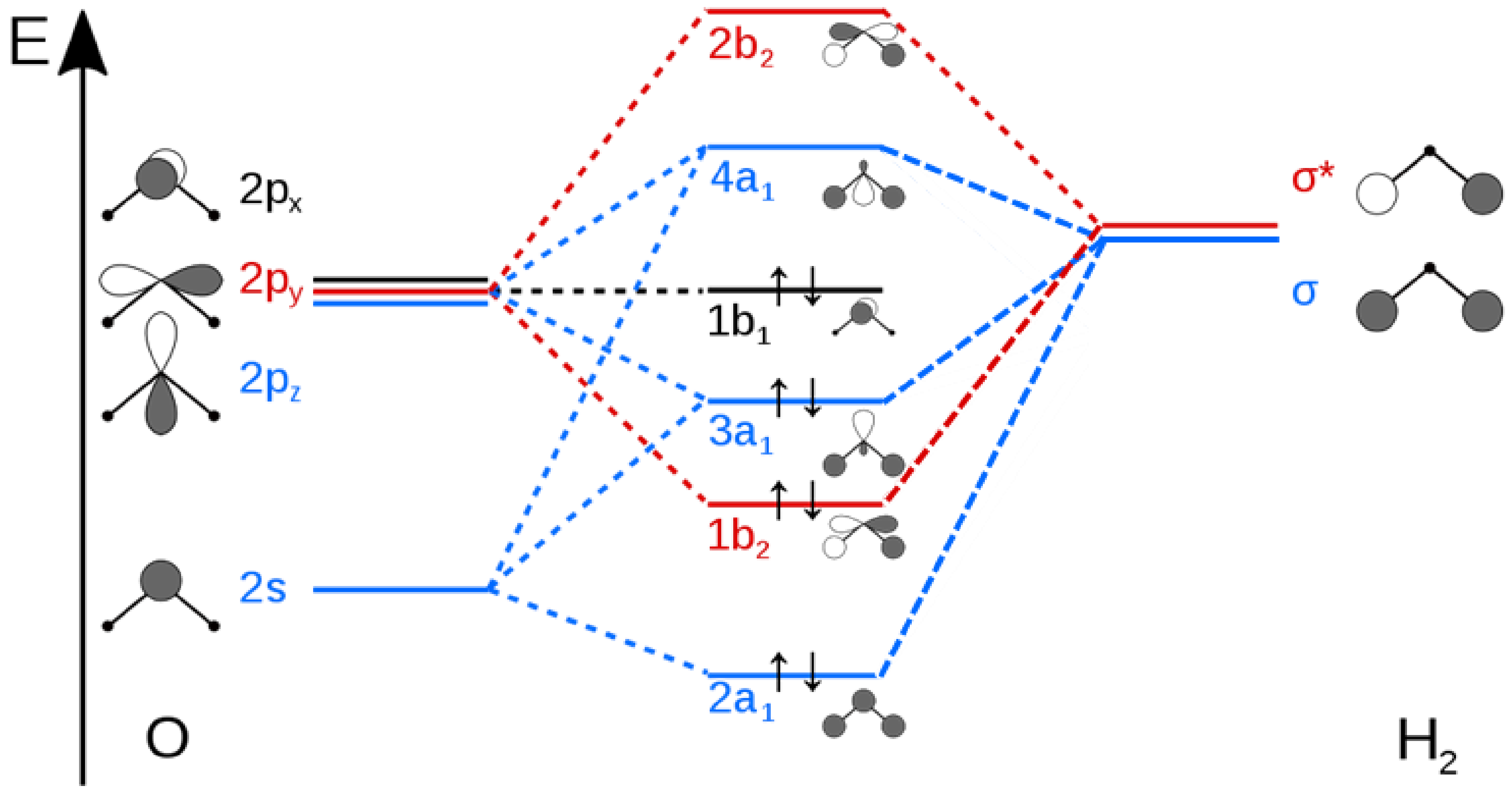
— 로버트 B. 러플린, 이덕환 옮김, 『새로운 우주』


SOLIDS and SURFACES
A Chemist's View of Bonding in Extended Structures
by
Roald Hoffmann
WILEY-VCH

(1988)



# Molecular Orbitals: LCAO-MO




(Image: https://commons.wikimedia.org/wiki/File:H2O-MO-Diagram.svg)

# Construction Rules for MO

In order to construct MOs from two basis orbitals,

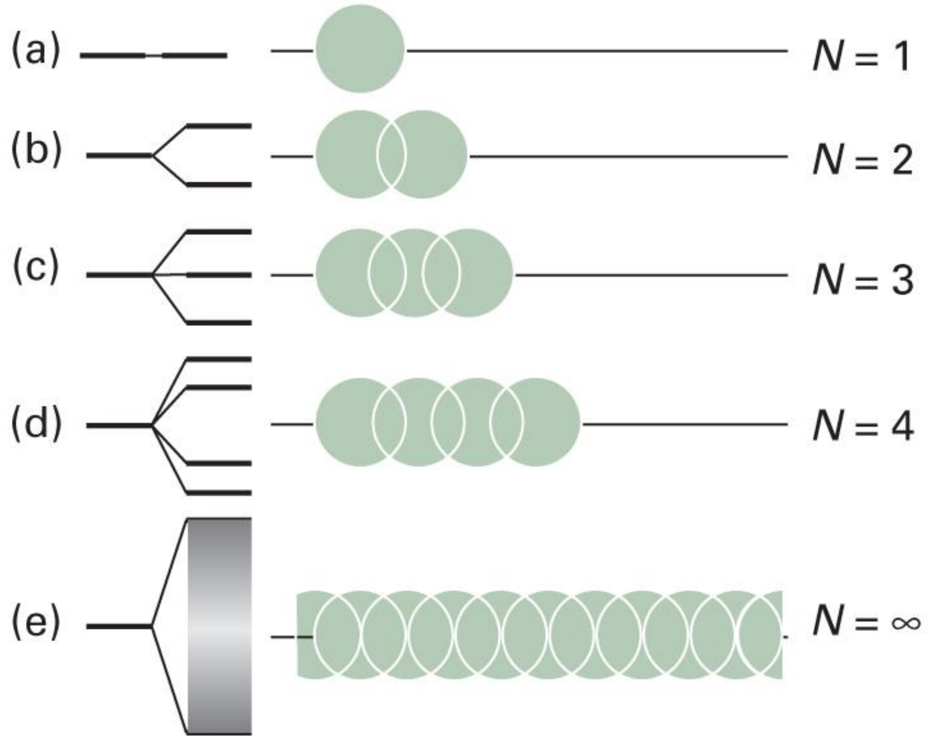
1. their overlap integral must be large enough

$$S_{ij} = \int \psi_i^* \psi_j \, \mathrm{d}\tau > 0$$

2. The energy levels must be close enough

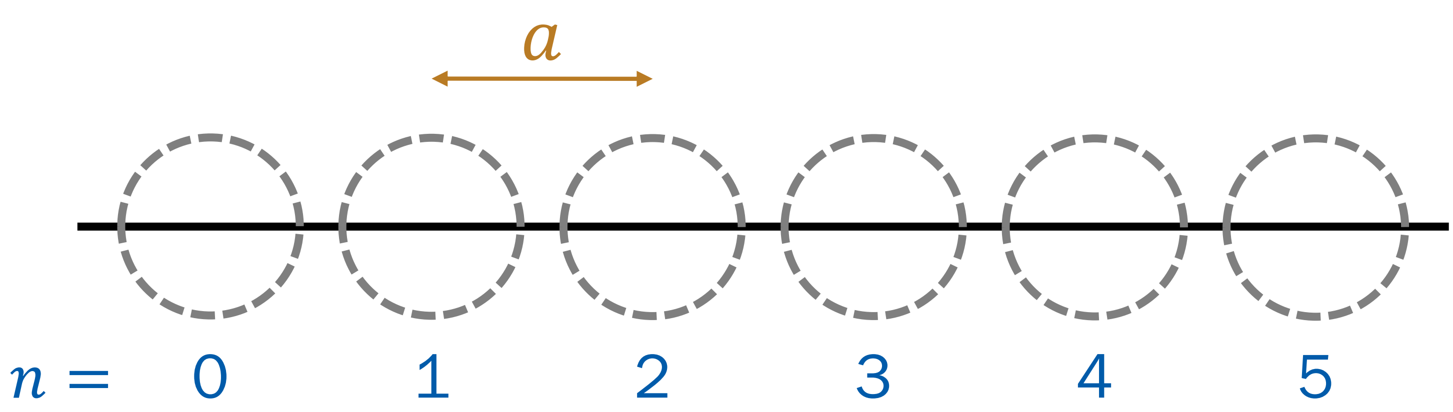
$$\psi_n^{(1)} = \sum_{m \neq n} \frac{\int \psi_m^{(0)*} \hat{H}' \psi_n^{(0)} \, \mathrm{d}\tau}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}$$

# Electronic Structures of Solids




(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 15C.6)

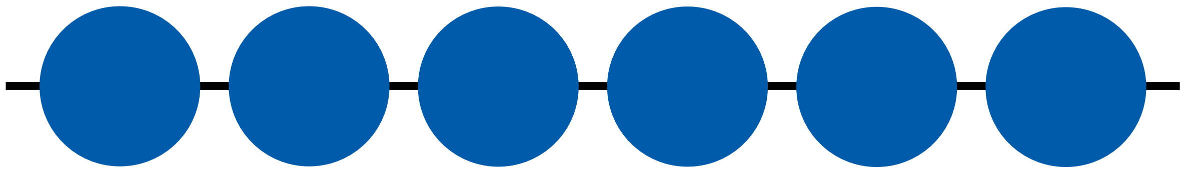
# Linear Combination of AOs



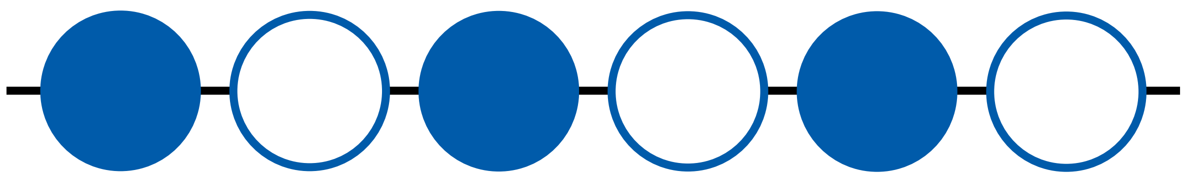

$$\phi_k = \sum_n e^{ikna} \psi_n$$

# Possible Combinations

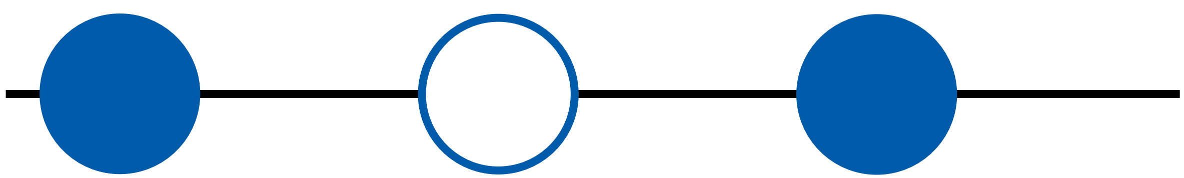
$$k = 0\colon \phi_0 = \sum_n \psi_n$$



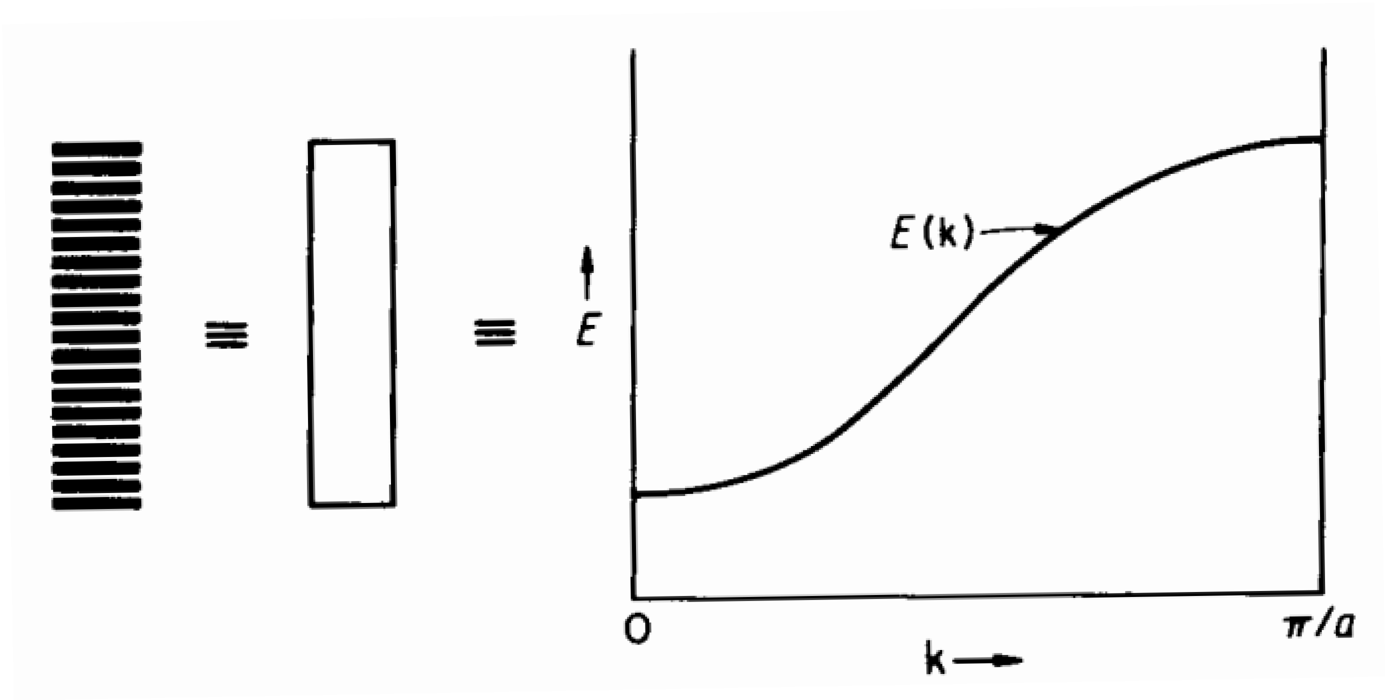
$$k = \pi/a\colon \phi_{\pi/a} = \sum_n (-1)^n \psi_n$$



$$k = \pi/2a\colon \phi_{\pi/2a} = \sum_n e^{in/2} \psi_n$$

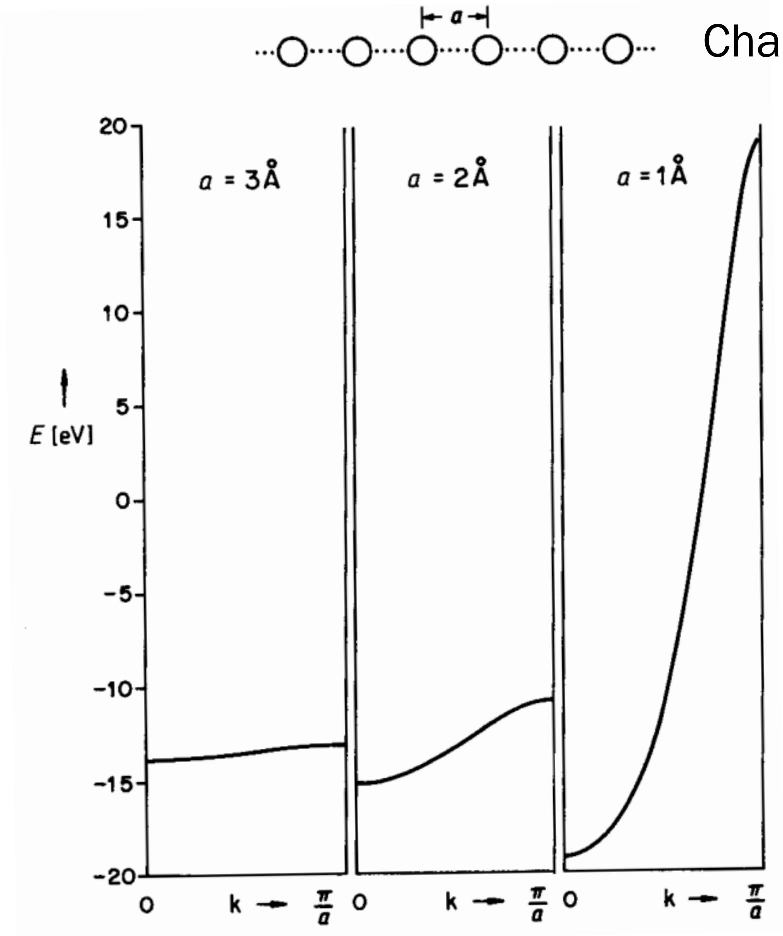


# “Band” and Dispersion Relation




(Image: Hoffmann, *Solids and Surfaces*, p. 7)

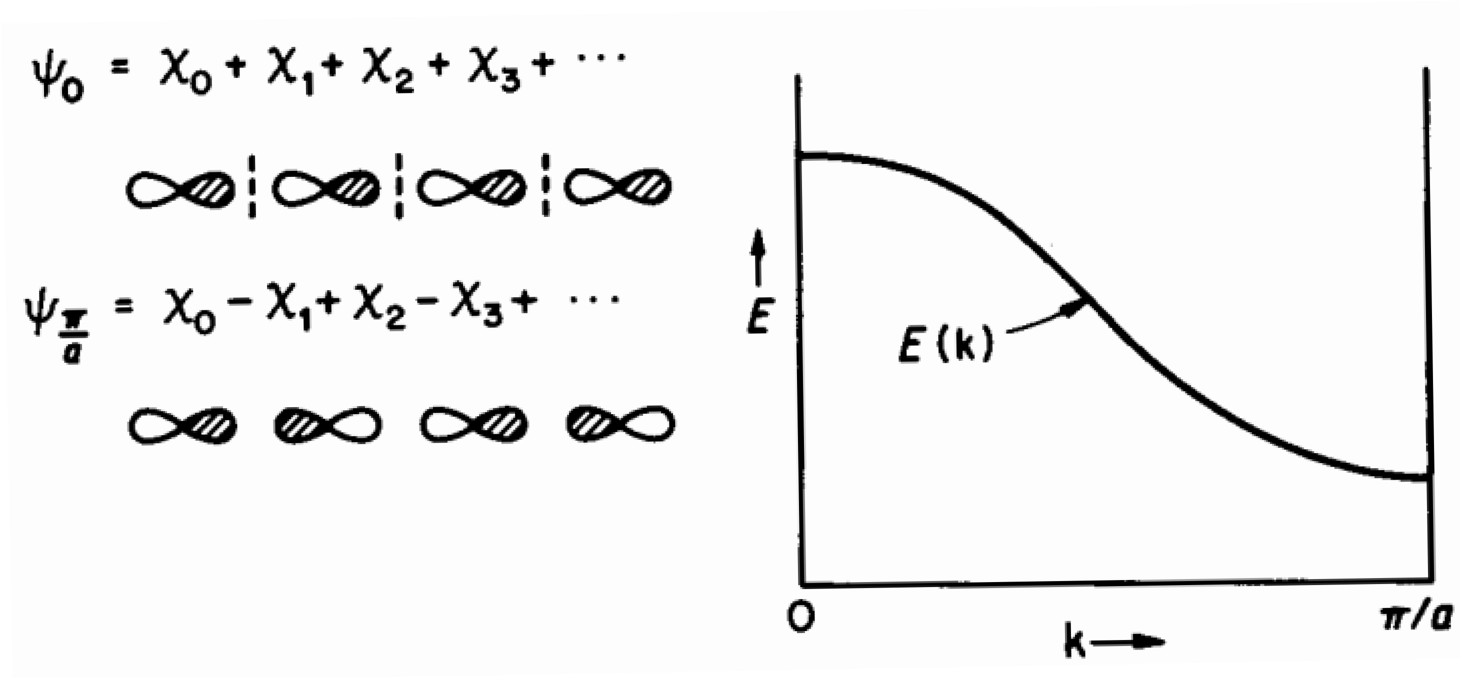
# “Band Width”




Chain of H atoms

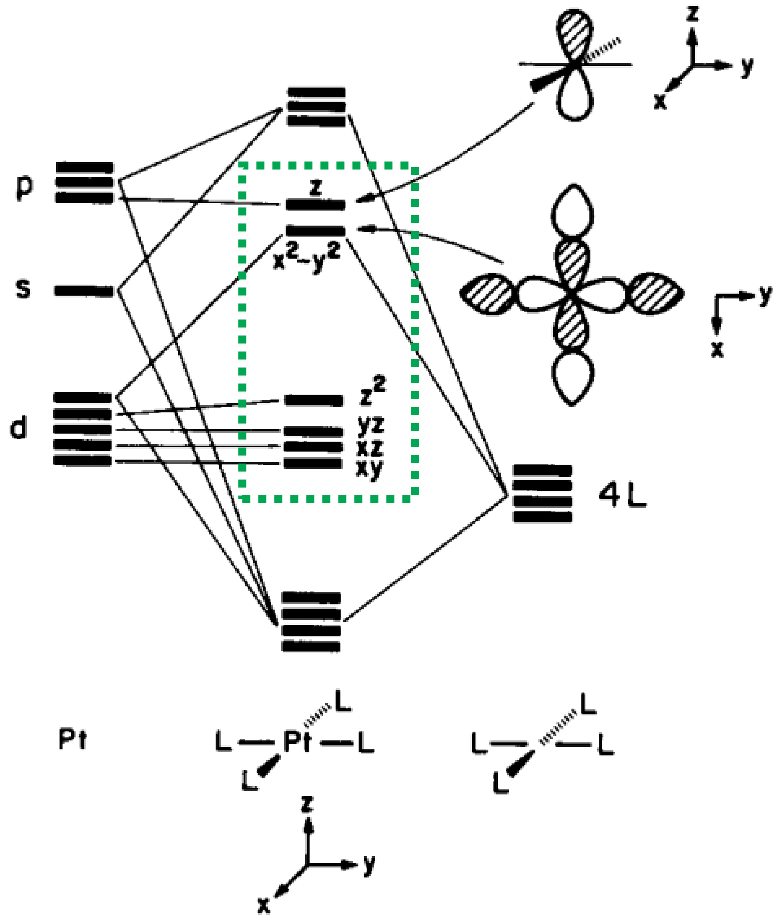
(Image: Hoffmann, *Solids and Surfaces*, Fig. 1)

# p Orbitals



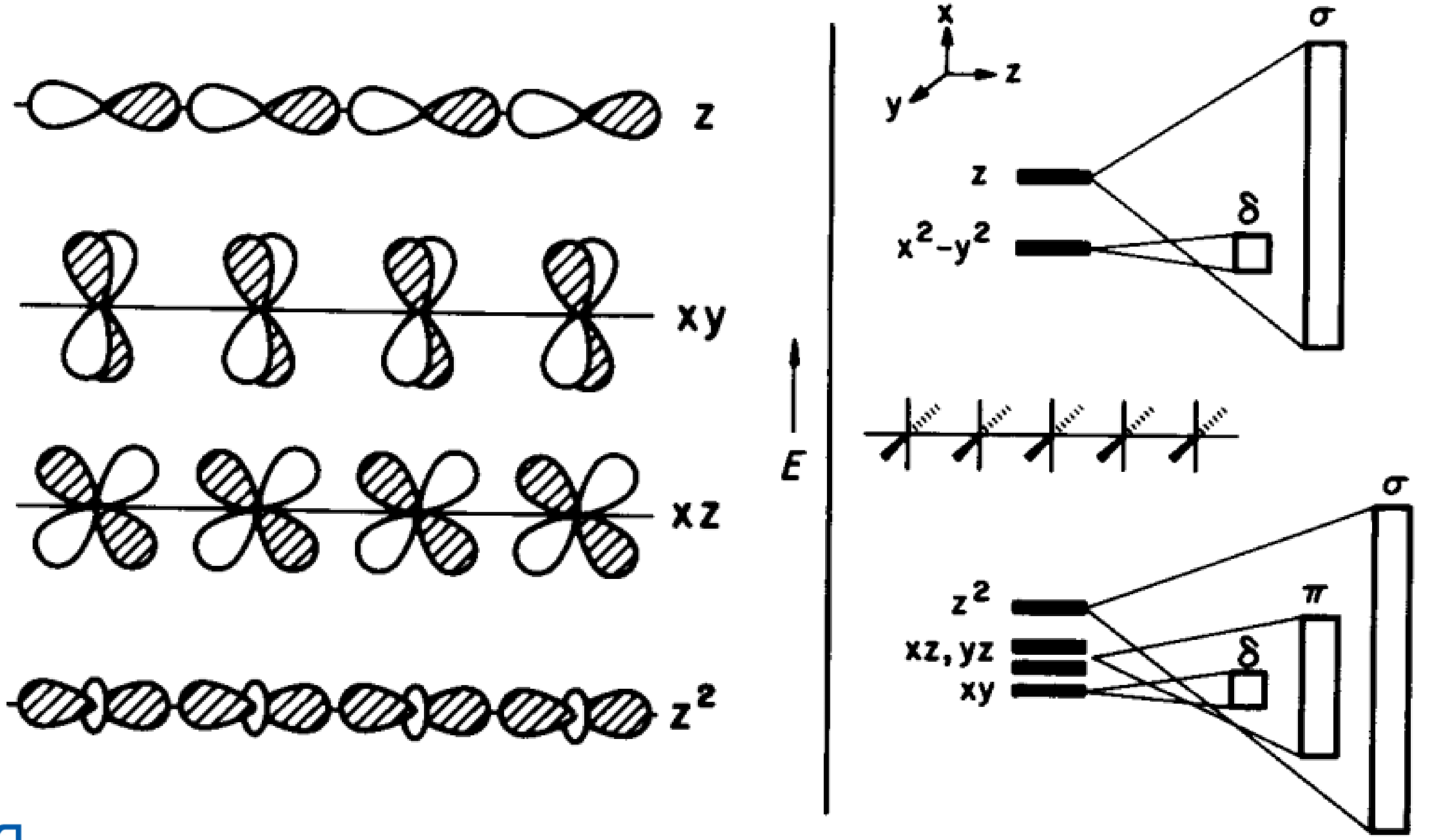

(Image: Hoffmann, *Solids and Surfaces*, p. 9)

# Example: $d^8$ $PtL_4$ Complex Chain



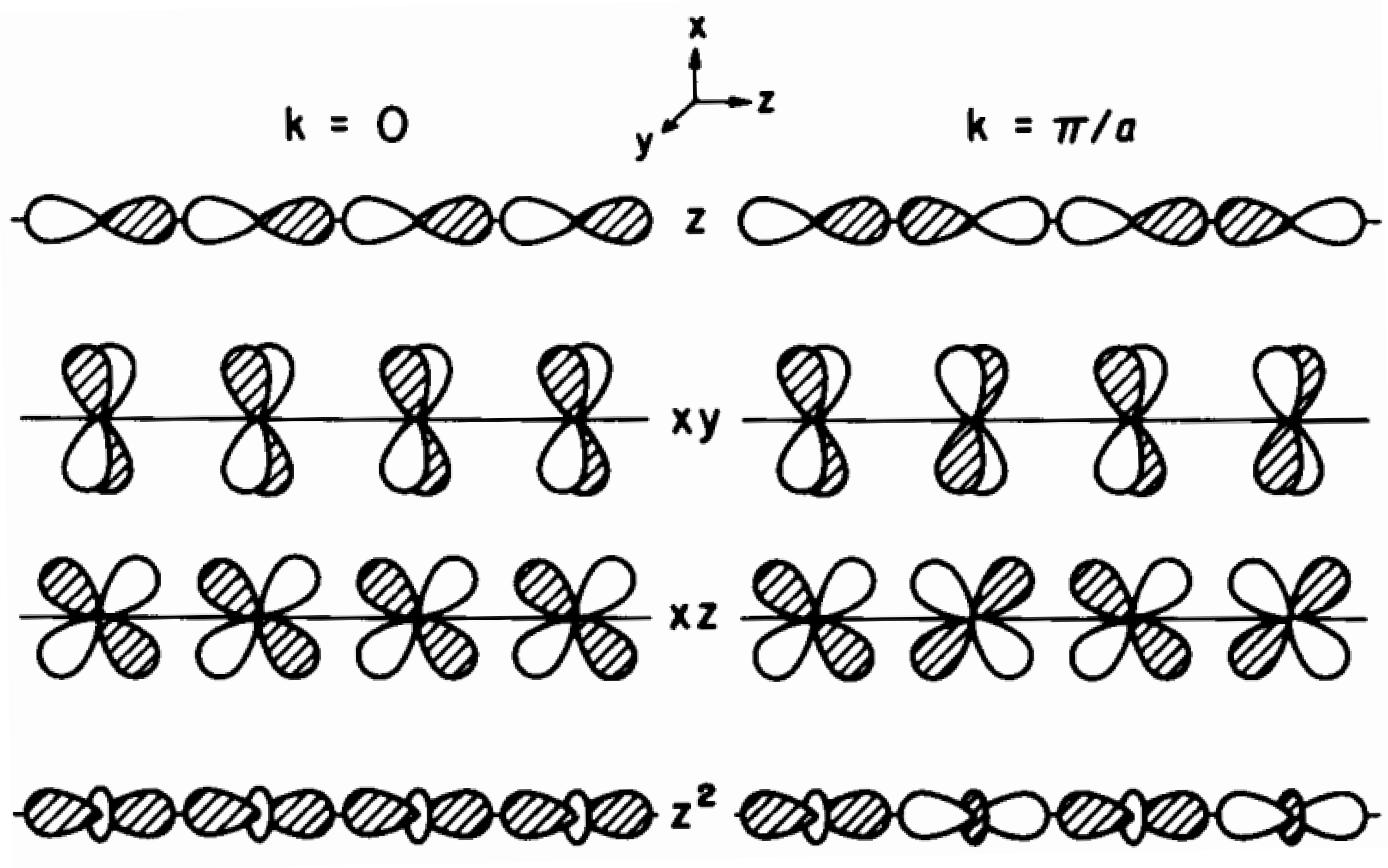

(Image: Hoffmann, *Solids and Surfaces*, Fig. 2)

# Bond Types: σ, π, δ, ...



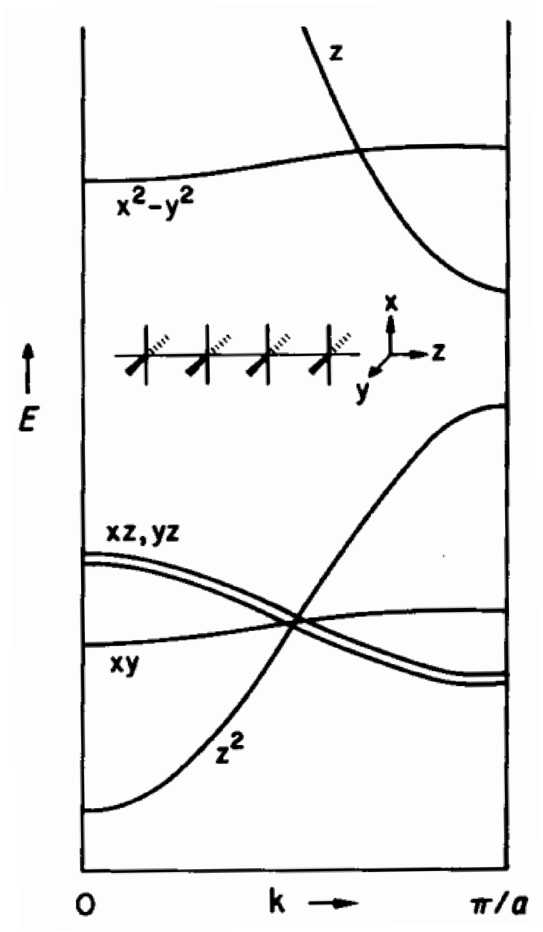

(Image: Hoffmann, *Solids and Surfaces*, pp. 12-13)

# Bonding vs. Antibonding



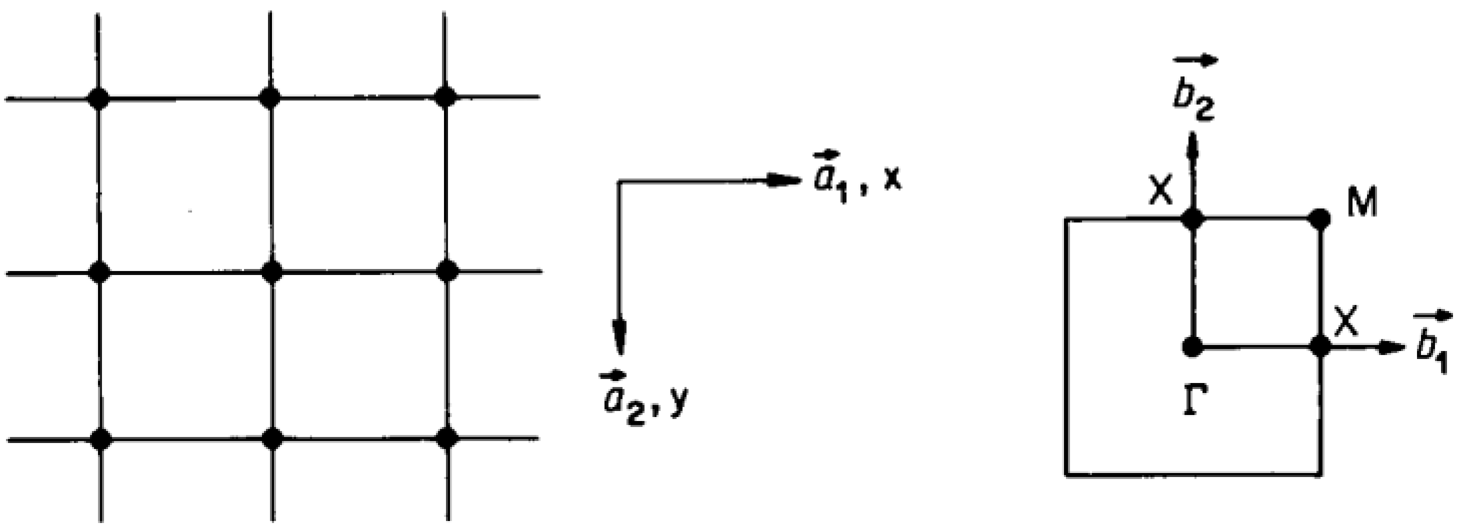

(Image: Hoffmann, *Solids and Surfaces*, p. 13)

# Finally




(Image: Hoffmann, *Solids and Surfaces*, p. 13)

# 2D: Square Lattice

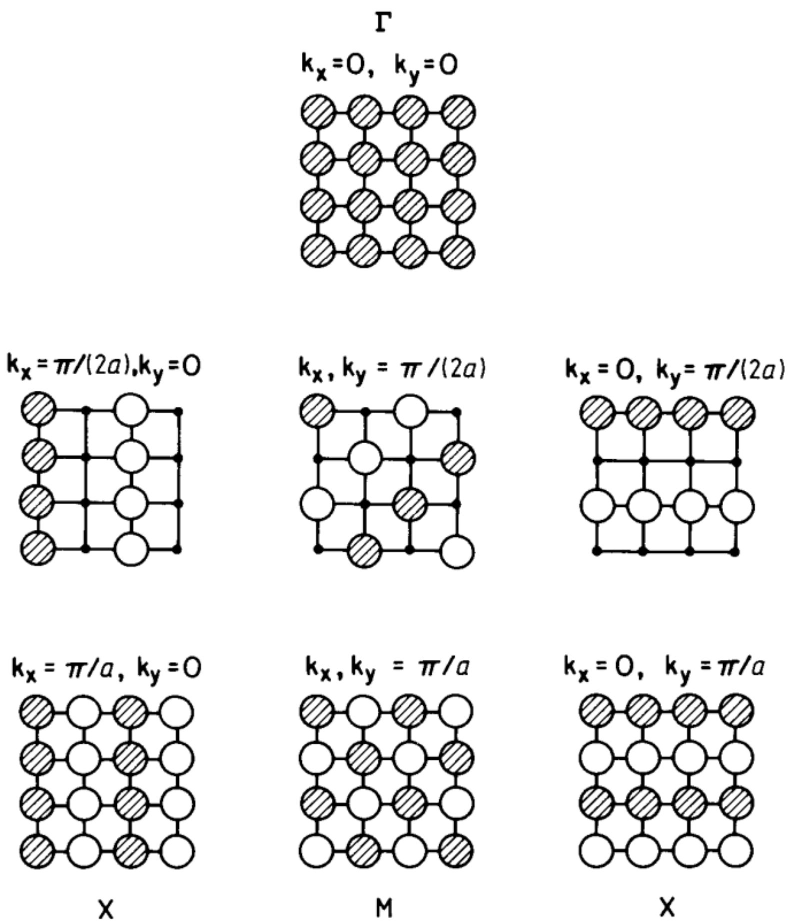


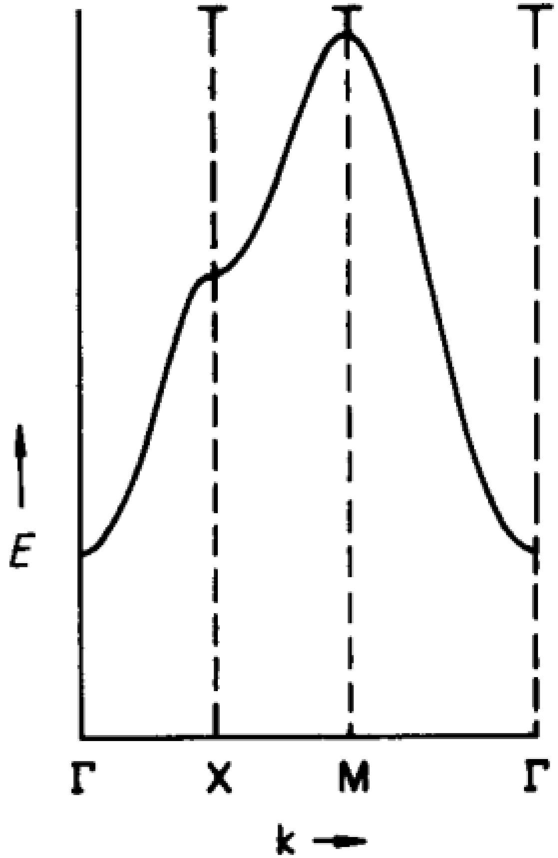

Real space

Reciprocal space

(Image: Hoffmann, *Solids and Surfaces*, pp. 16-17)

# Combinations of AOs

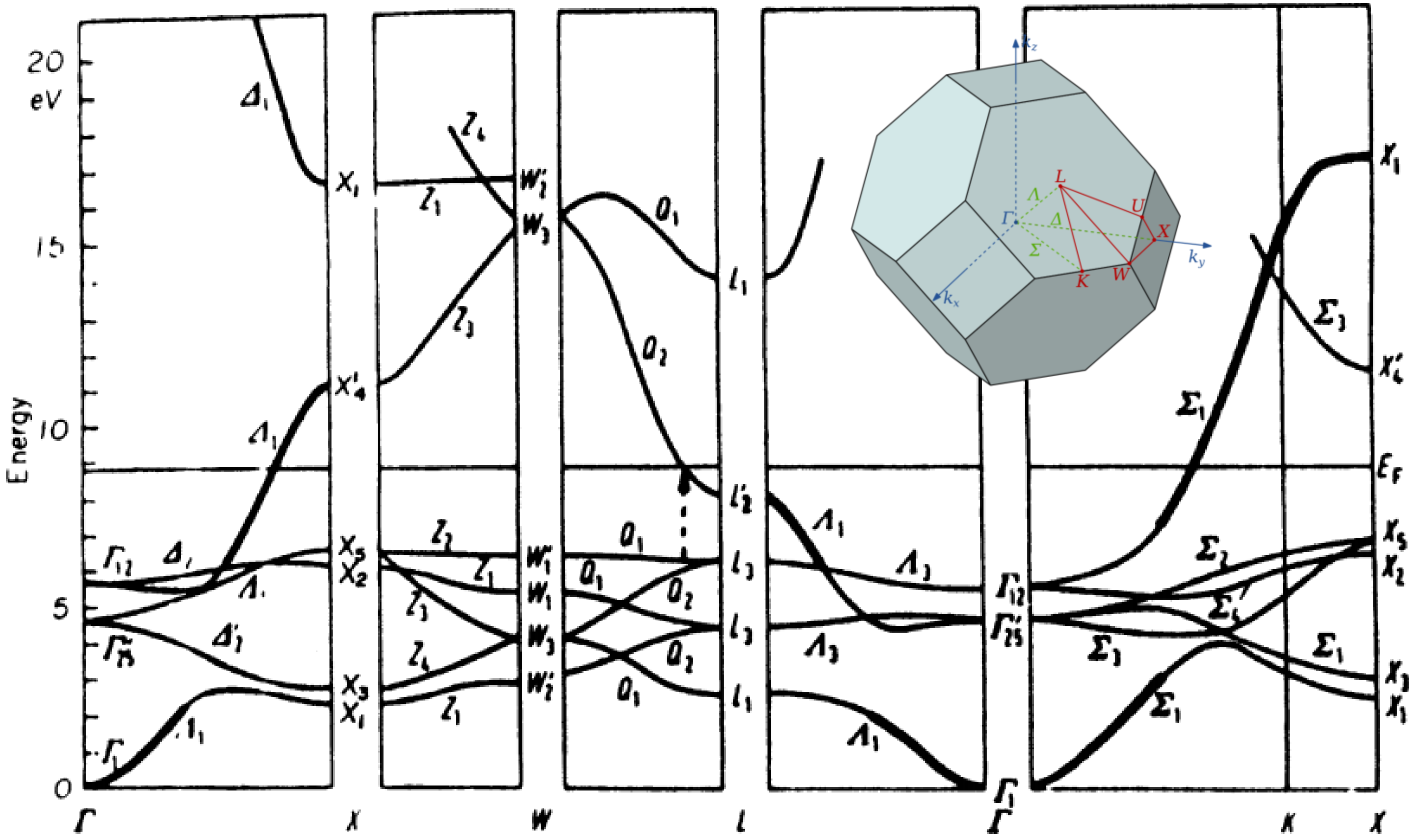



(Image: Hoffmann, *Solids and Surfaces*, pp. 17-18)

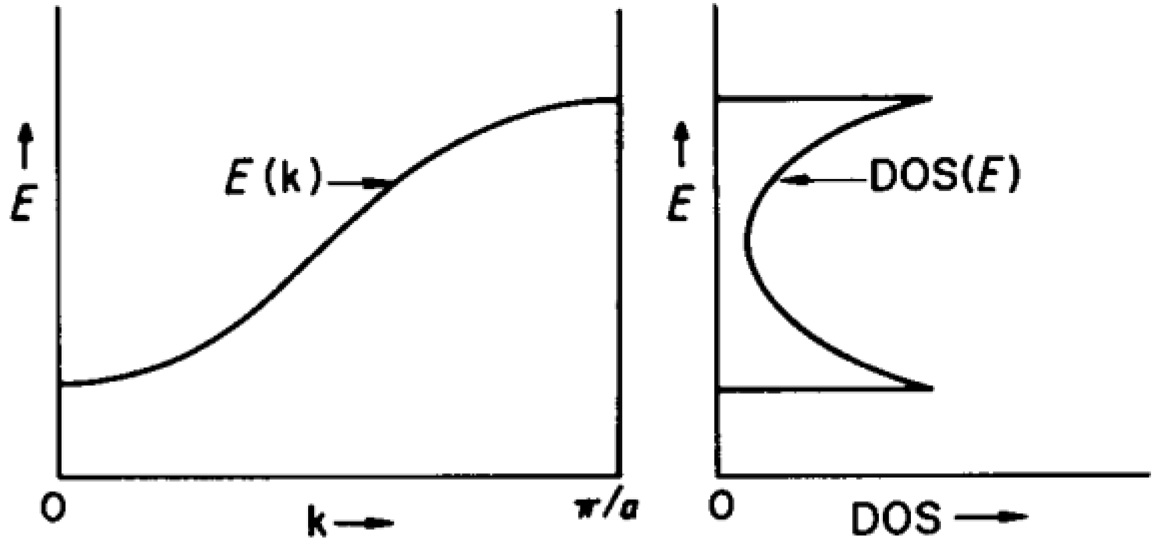
# Dispersion Relation in 3D

Cu (fcc)



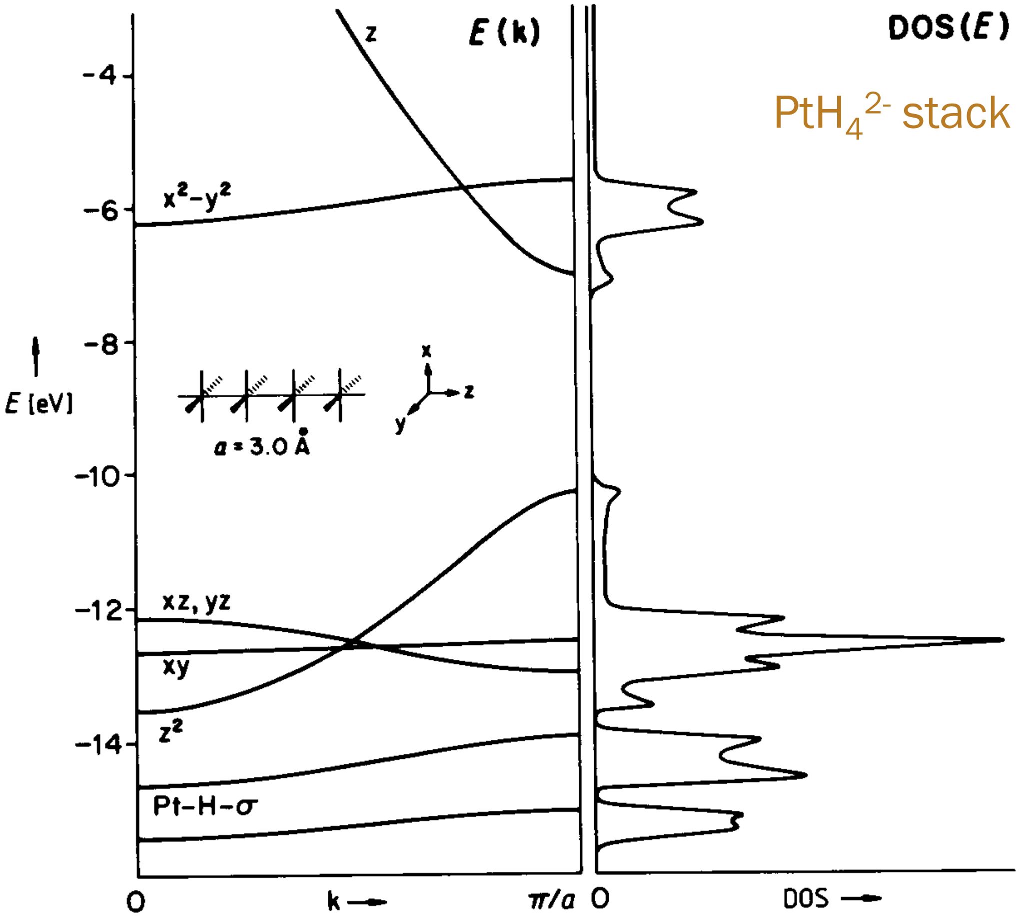

(Image: Segall, *Phys Rev* (1962), Fig. 6)

# Density of States (DOS)



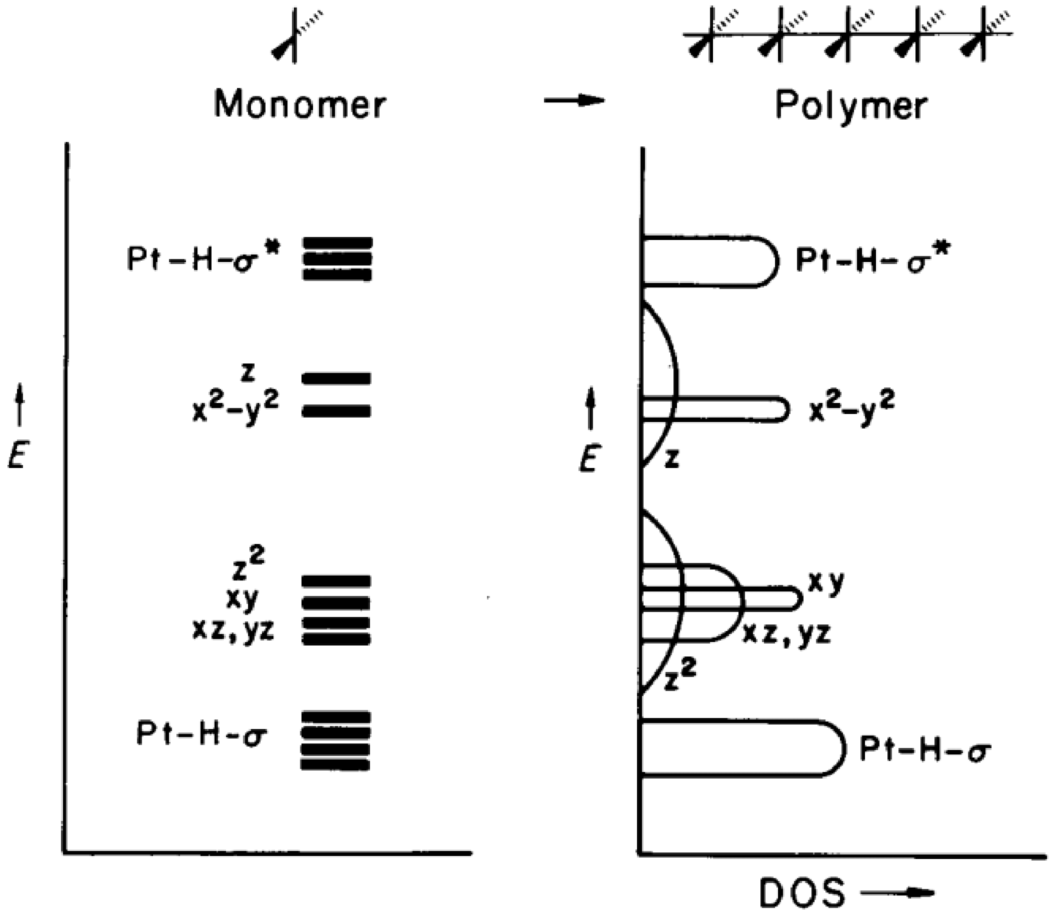

(Image: Hoffmann, *Solids and Surfaces*, p. 27)

# DOS of Previous Example




(Image: Hoffmann, *Solids and Surfaces*, p. 28)

# Qualitatively



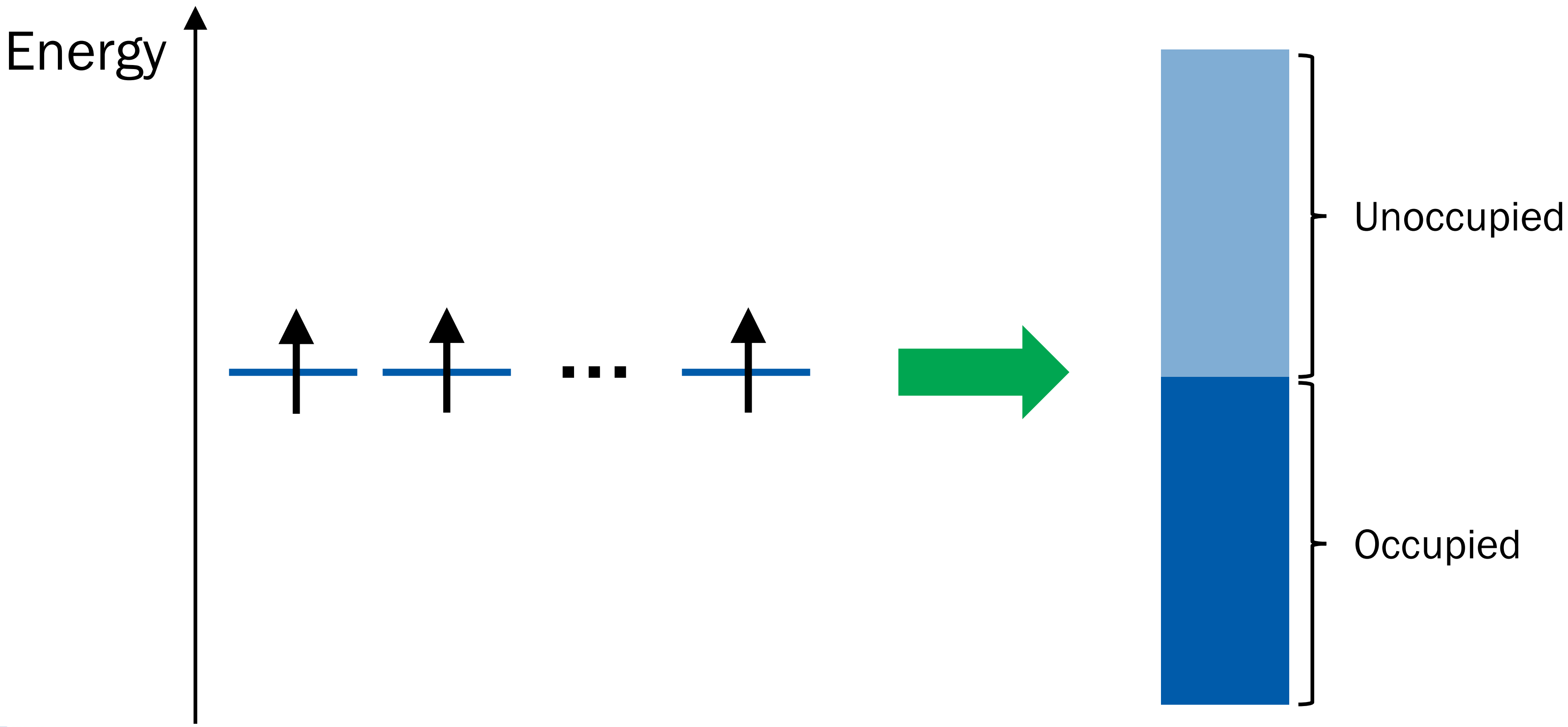

(Image: Hoffmann, *Solids and Surfaces*, p. 29)

# Where are the electrons?

Aufbau Principle!

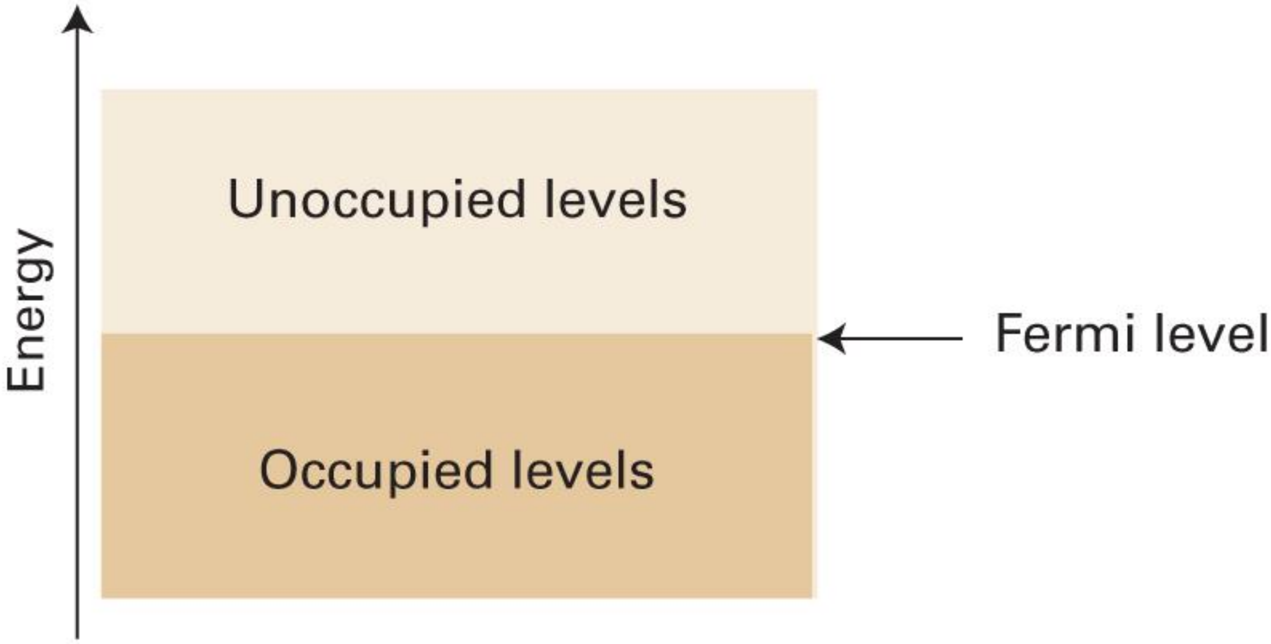
# Band Structure

$(T = 0)$



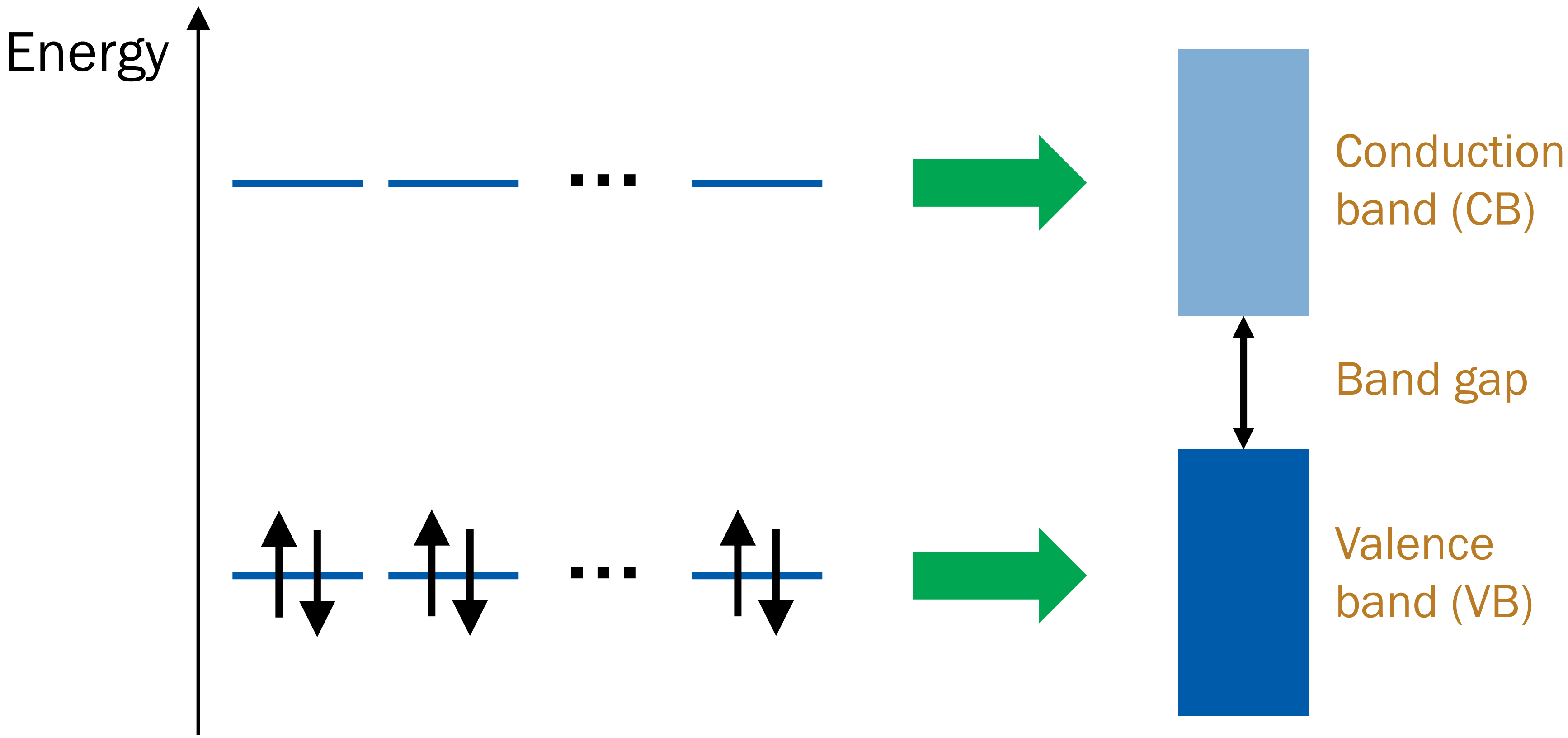
# Fermi Level

$(T = 0)$

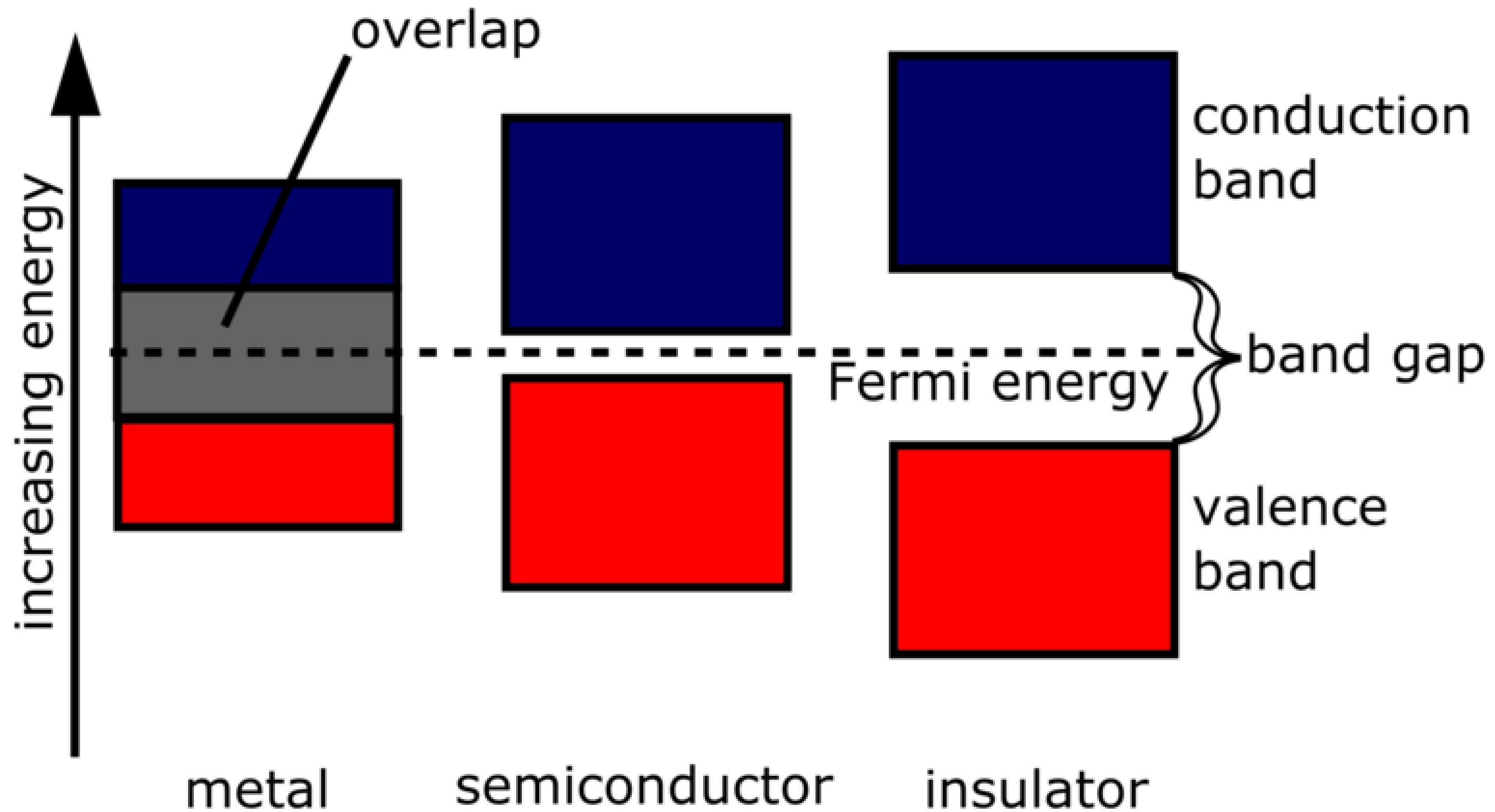



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 15C.8)

# Multiple AOs

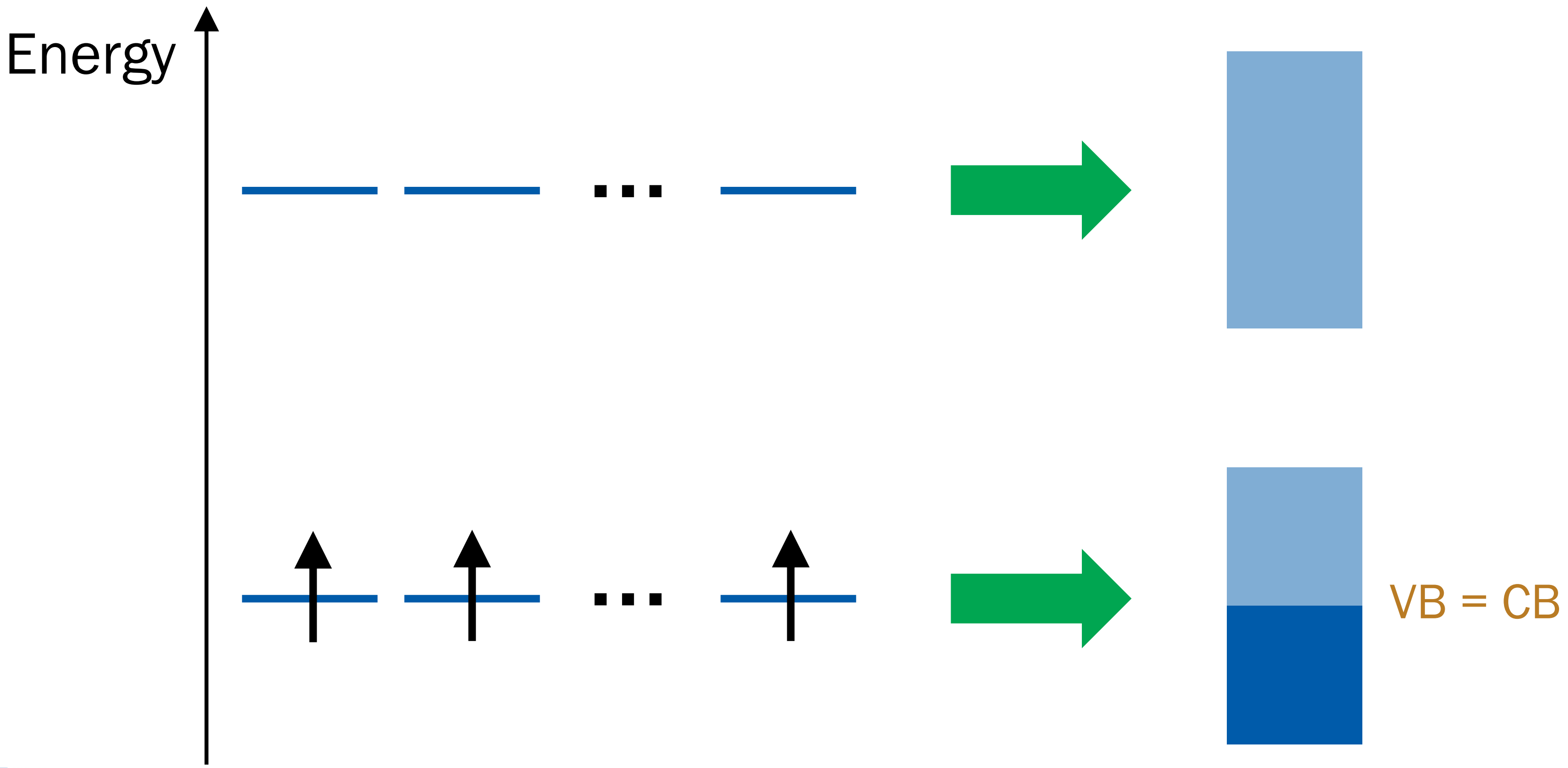


# Band Gap Matters

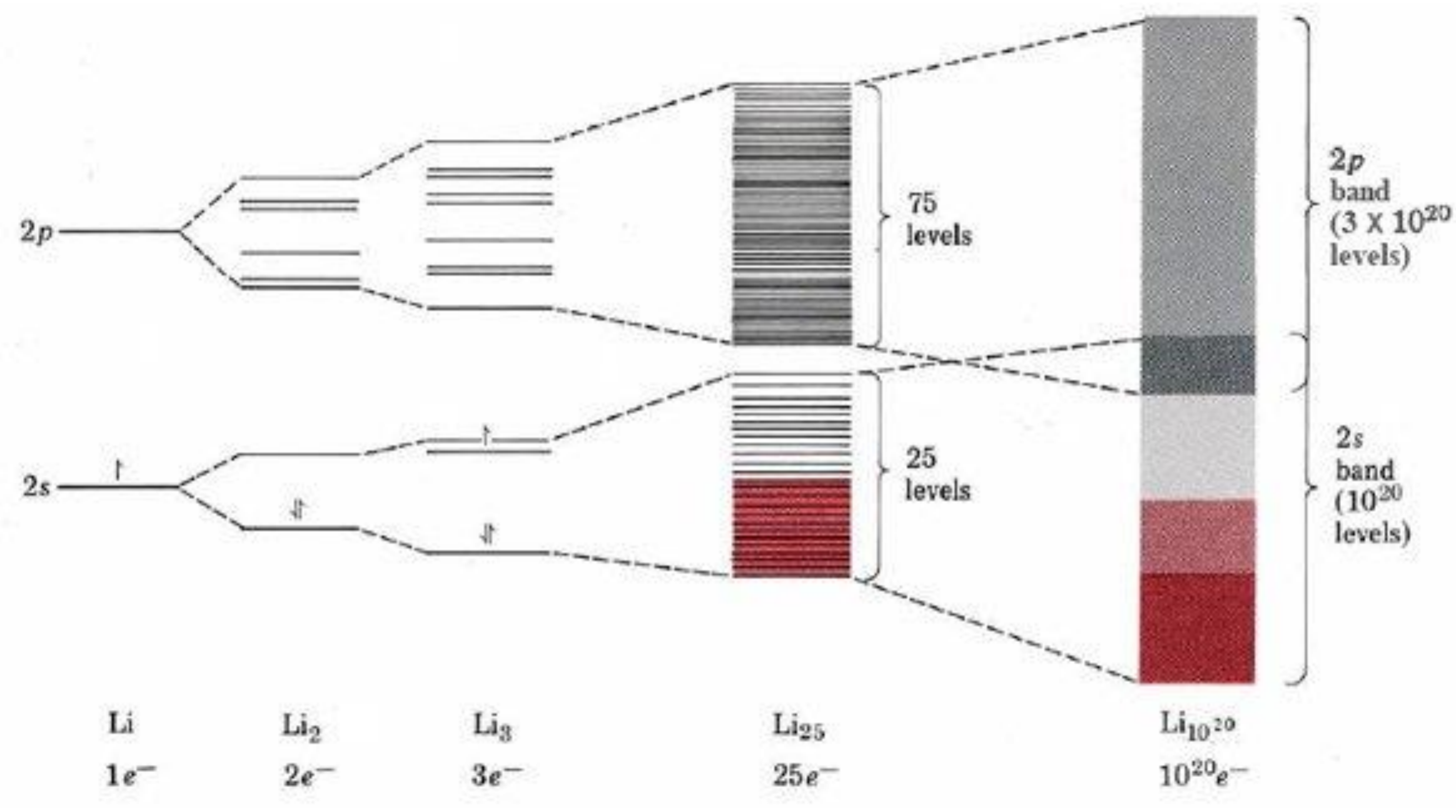



(Image: https://energyeducation.ca/encyclopedia/Band_gap)

# Single Electrons

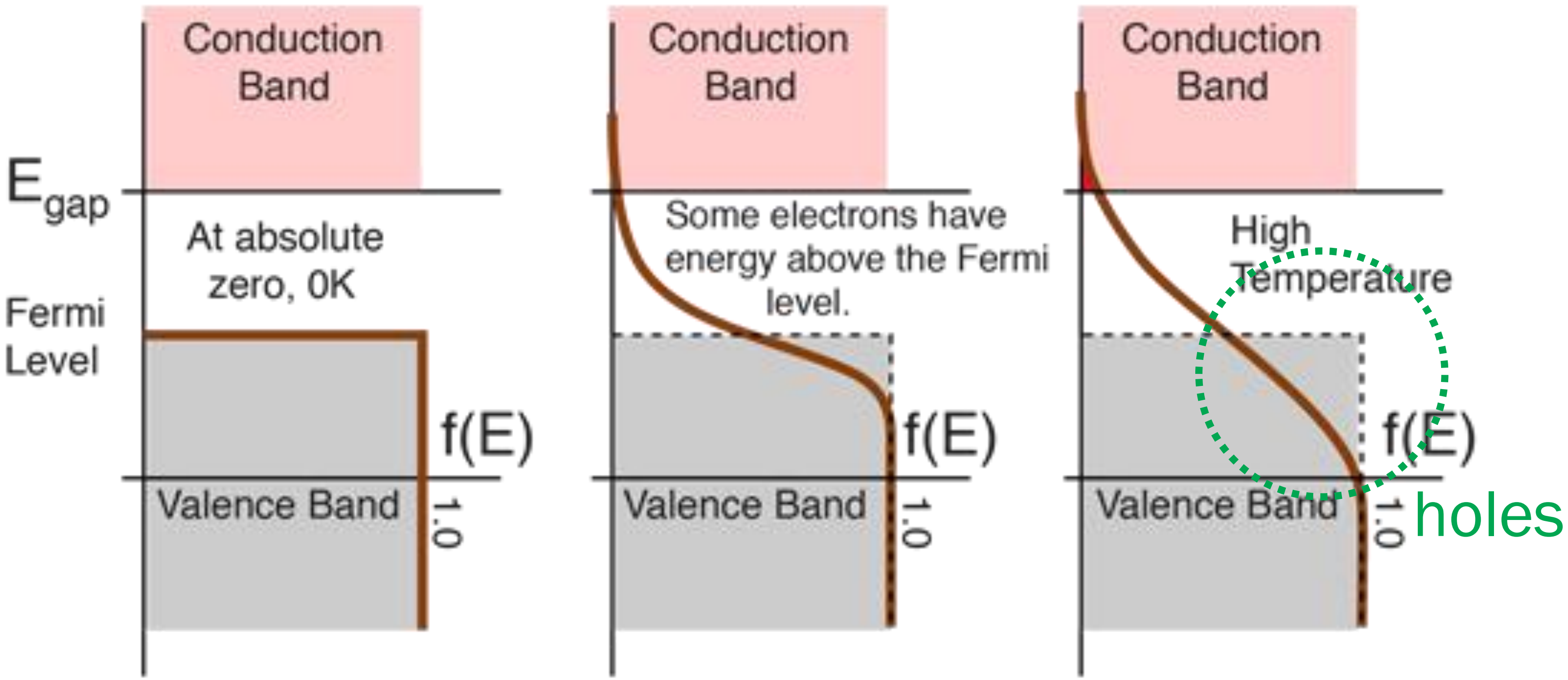


# Case of Lithium



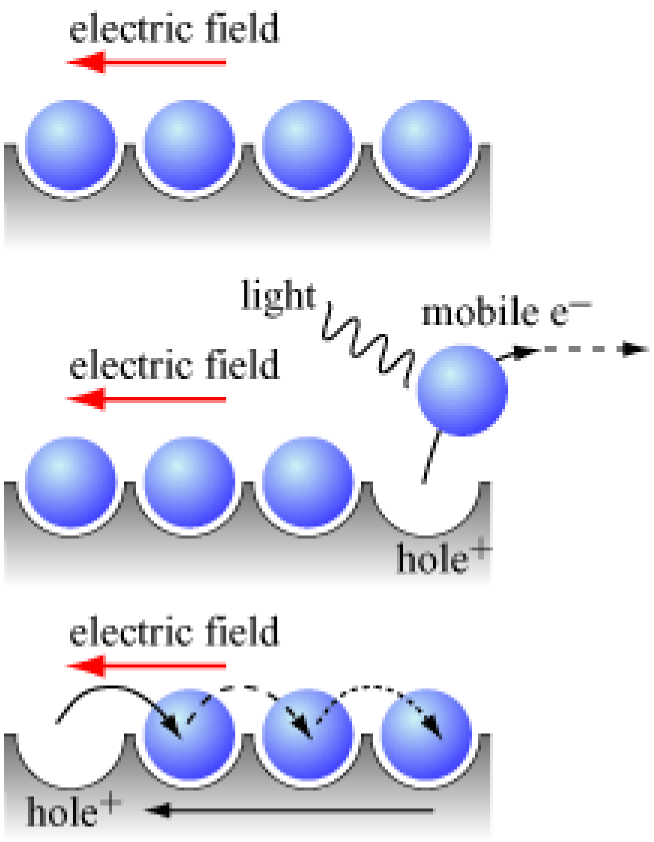

(Image: https://www.toppr.com/ask/content/concept/band-theory-254488/)

# At Finite Temperature



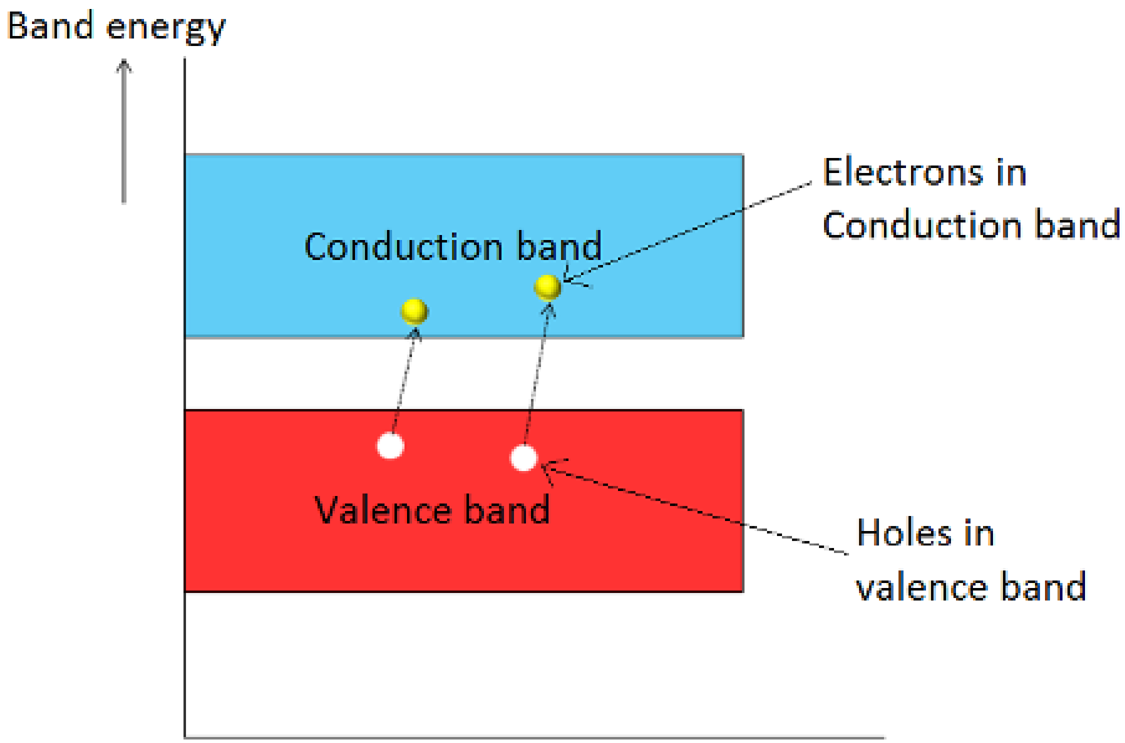

(Image: http://hyperphysics.phy-astr.gsu.edu/hbase/Solids/Fermi.html)

# Holes: Quasiparticles



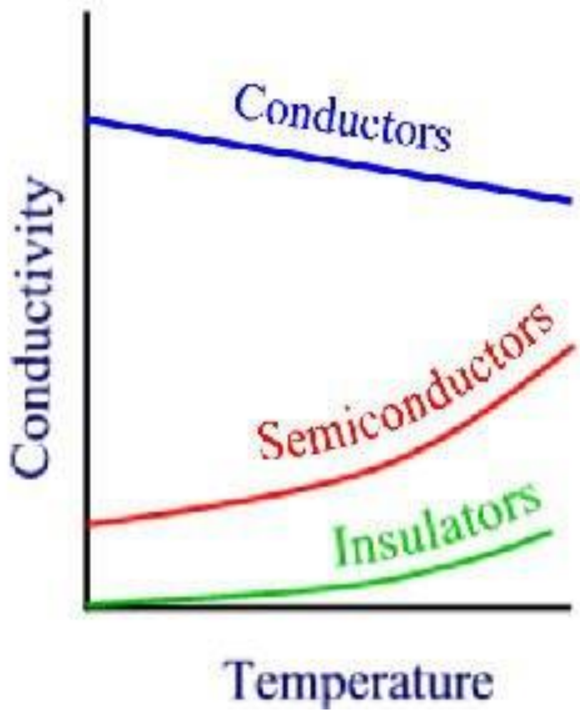

(Image: https://physics.stackexchange.com/questions/15640/what-are-electron-holes-in-semiconductors)

# Semiconductors at Finite $T$



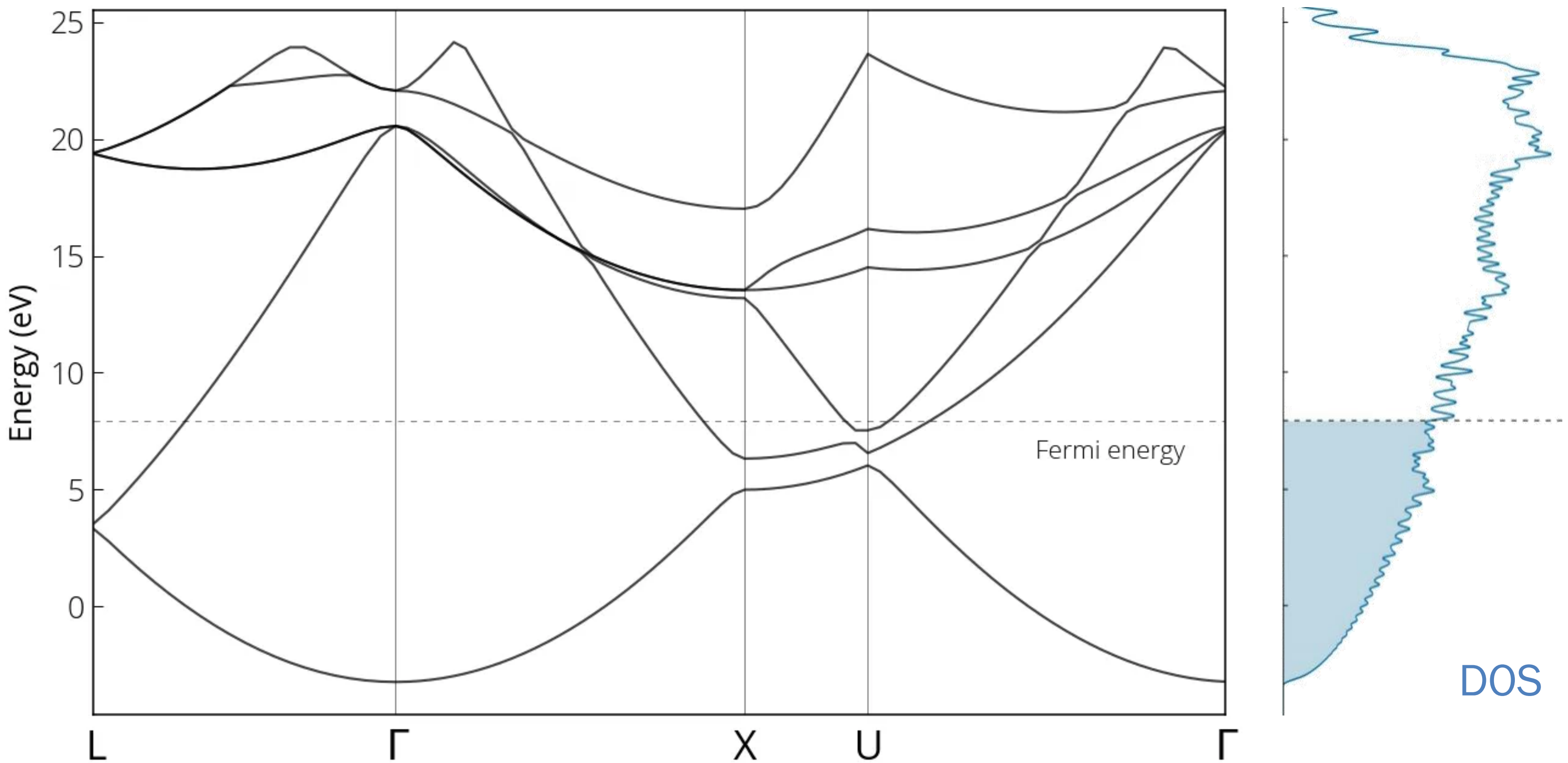

(Image: https://www.physics-and-radio-electronics.com/electronic-devices-and-circuits/semiconductor/hole.html)

# Temperature and Conductivity



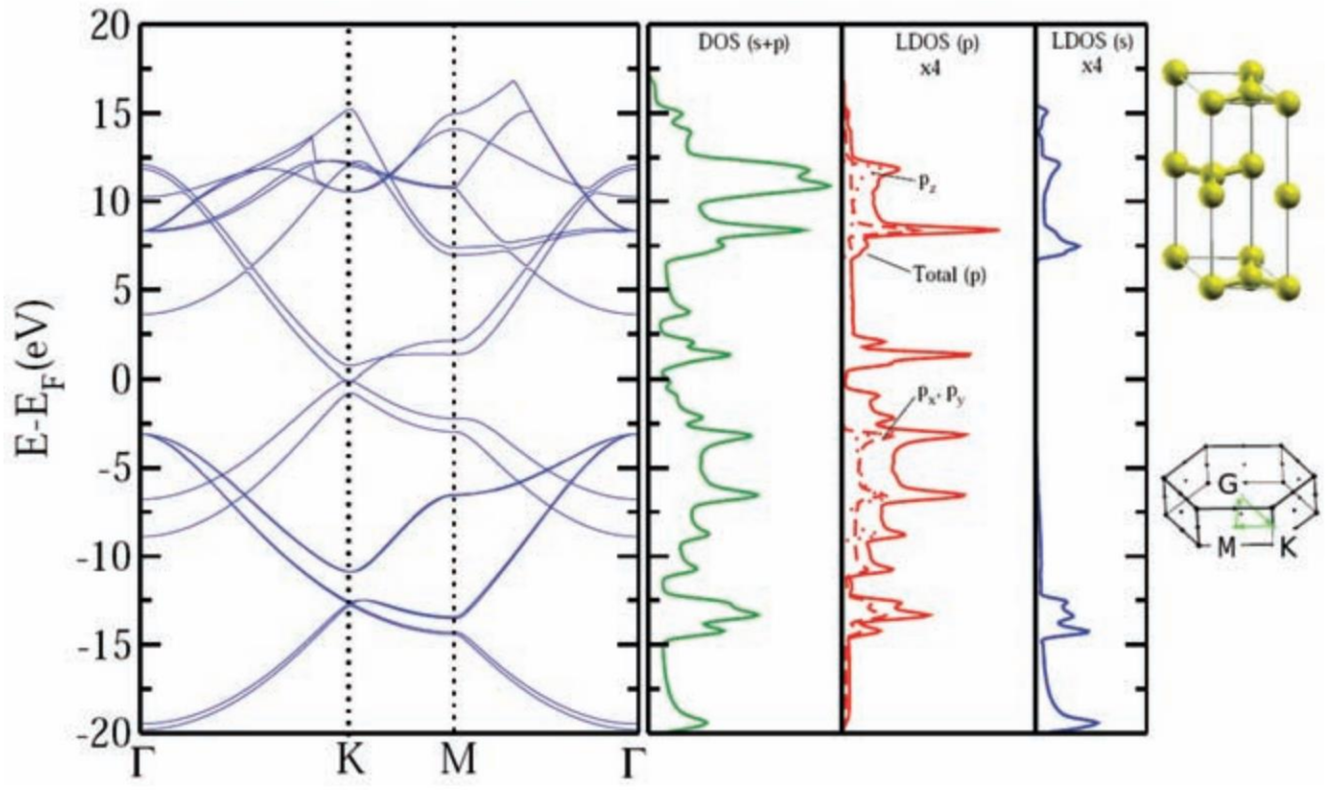

(Image: https://atlas-scientific.com/blog/why-does-conductivity-increase-with-temperature-in-semiconductors/)

# Example: Aluminum



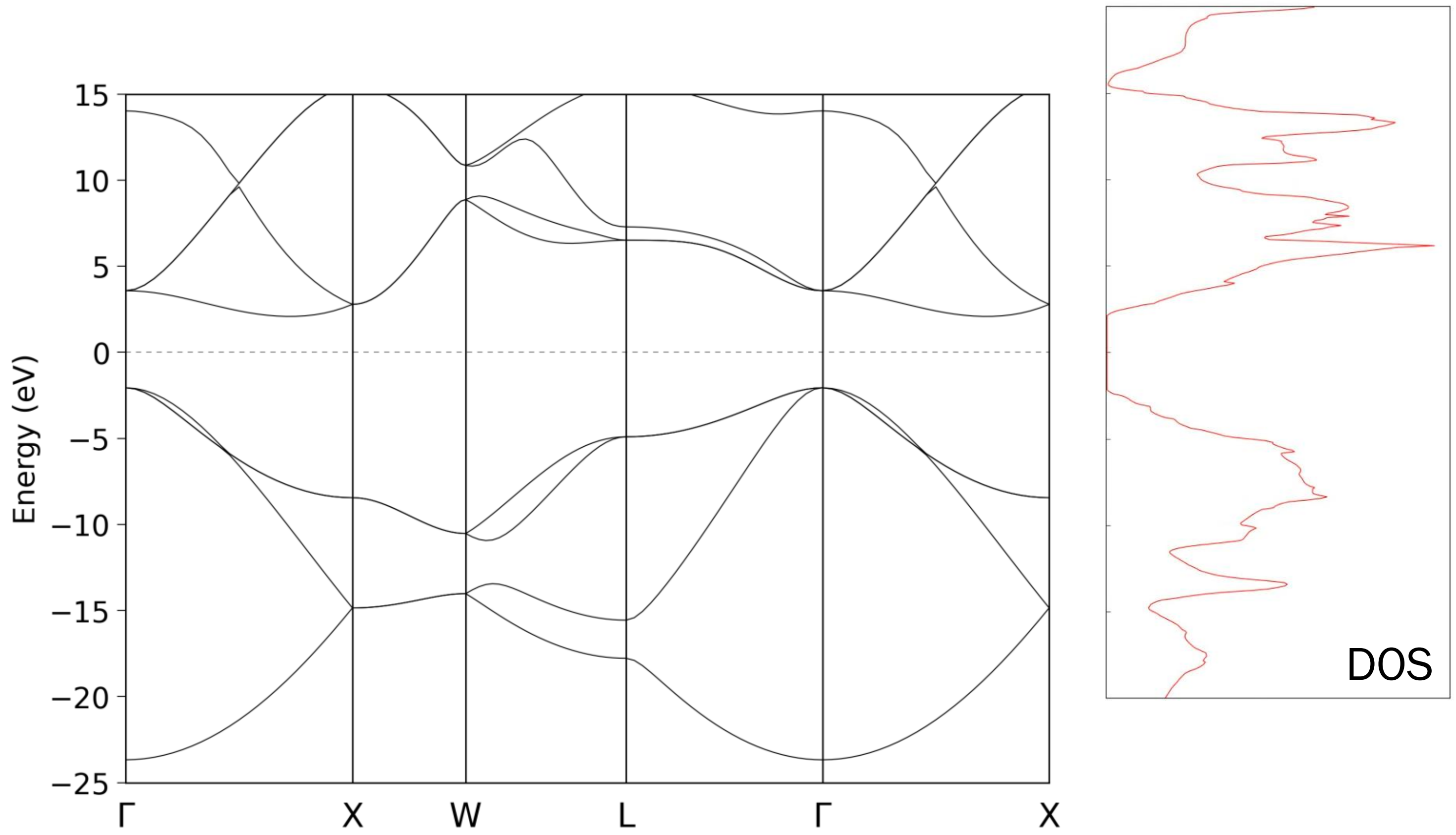

(Image: https://pranabdas.github.io/espresso/hands-on/aluminum/)

# Example: Graphite




(Image: Kaloni and Mukherjee, *Mod Phys Lett B* (2011), Fig. 6)

# Example: Diamond




(Image: https://pranabdas.github.io/openmx/diamond/)